S1 File. Supporting information.

Shared architecture, disease-context-dependent direction: a harmonised cross-disease computational analysis of translation-related transcriptional programs in hepatocellular carcinoma and heart failure

Figures S1–S5 · Tables S1–S40

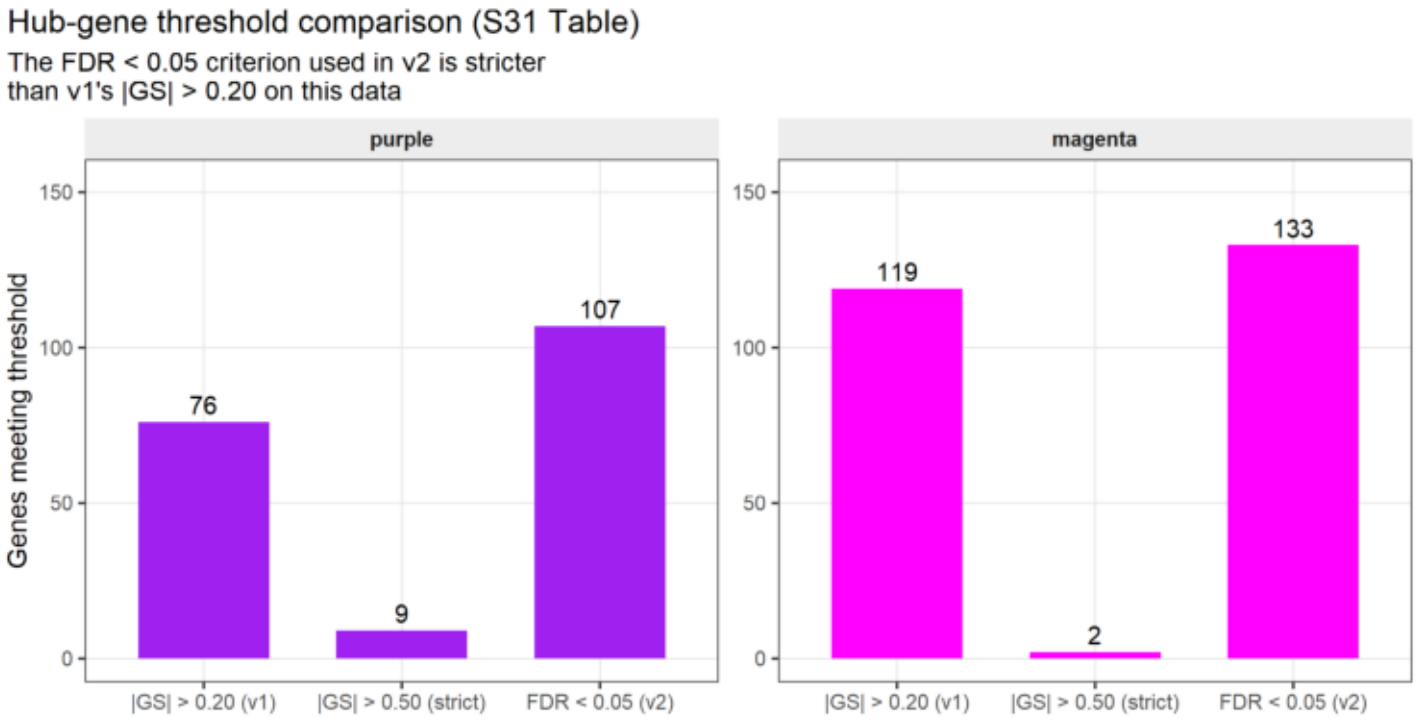
Every figure and table in this supplement is generated by the frozen pipeline under Code/Revision_v2/ from public data. The configuration hash recorded in S1 Table identifies the exact settings used.

Every table is also supplied complete as a machine-readable CSV alongside this document, and the CSV, not this PDF, is the authoritative full text. A table too large to print within its page budget is shown as a bounded preview that states its full row and column counts and names the CSV carrying the rest, and the rows the text cites are printed first. Long cell values are shortened with "..." in the printed preview only; the CSV holds them whole.

Generated by Code/Revision_v2/R19_build_SI_pdf.py

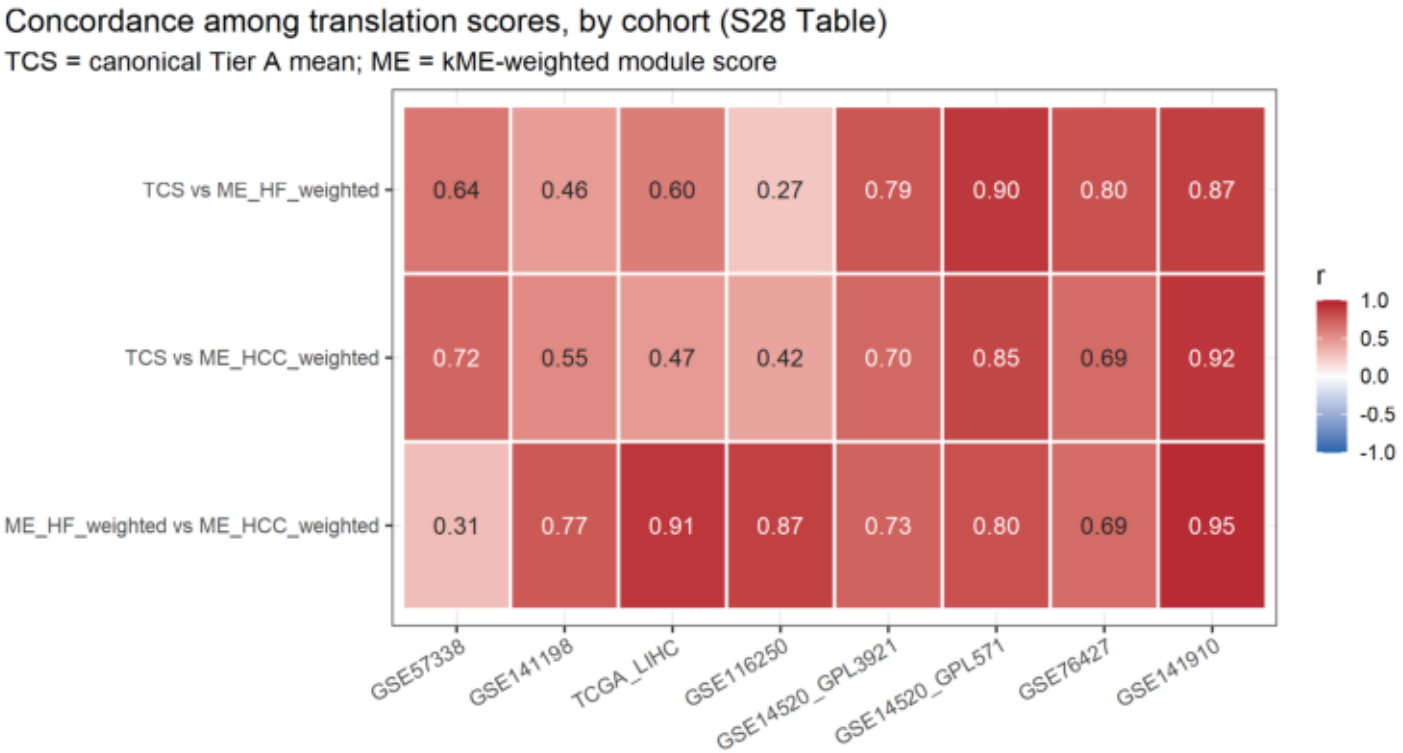
S1 Fig.

Hub-gene threshold comparison for the two designated translation modules (S31 Table). Gene counts meeting v1's |GS| > 0.20, a stricter |GS| > 0.50, and the FDR < 0.05 criterion adopted in v2. The criteria are nested on these data — every gene passing |GS| > 0.50 also passes FDR < 0.05 — and the counts are 76, 9 and 107 for HF purple and 119, 2 and 133 for HCC magenta. |GS| > 0.50 is therefore the strictest of the three and FDR < 0.05 the most permissive; the FDR criterion was adopted for its calibration to sample size rather than for added stringency.



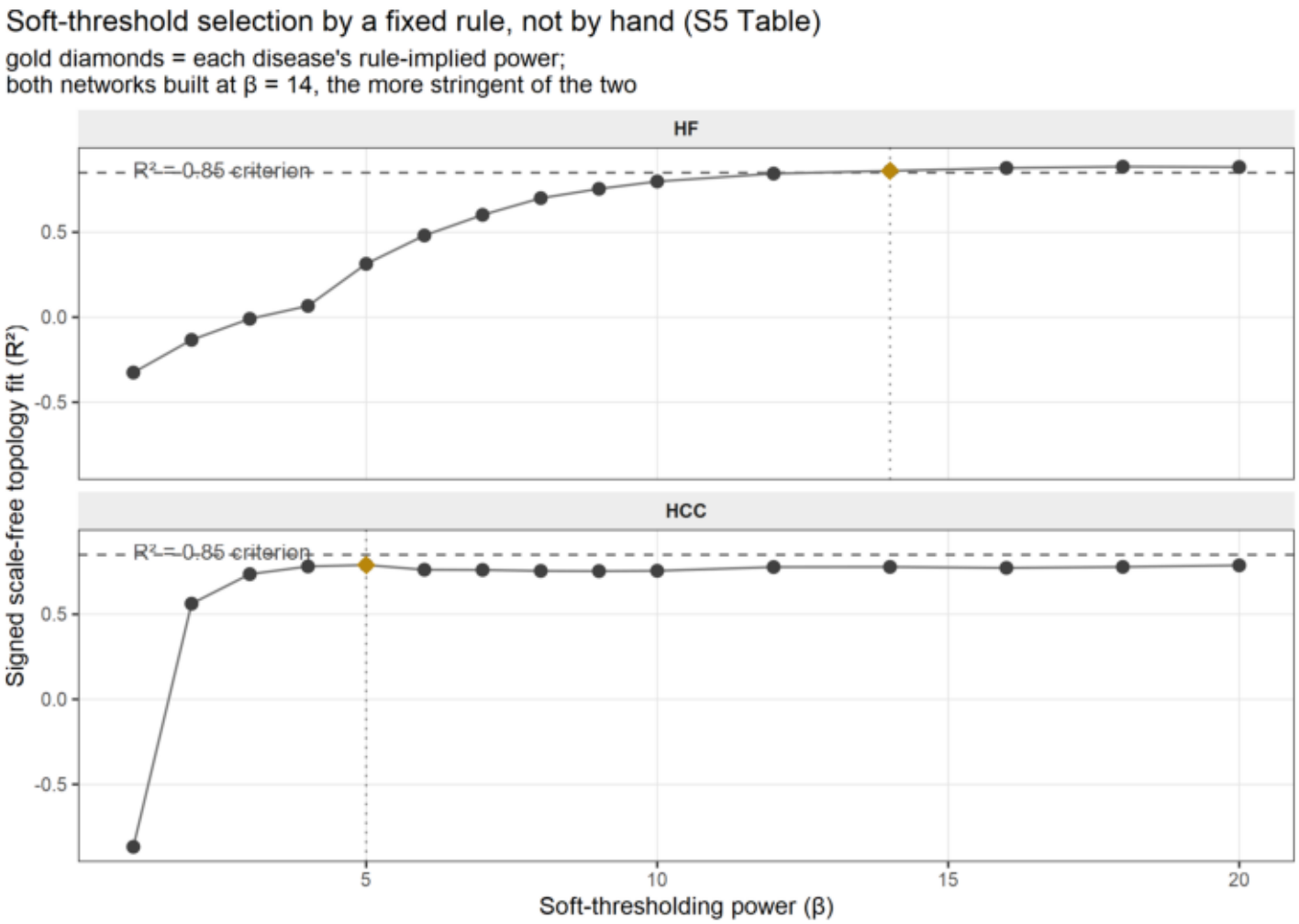
S2 Fig.

Cross-cohort effect-size concordance for the translation score. Module eigengene scores for the HF purple and HCC magenta modules, and the canonical translation score, computed in all eight cohorts and compared pairwise.



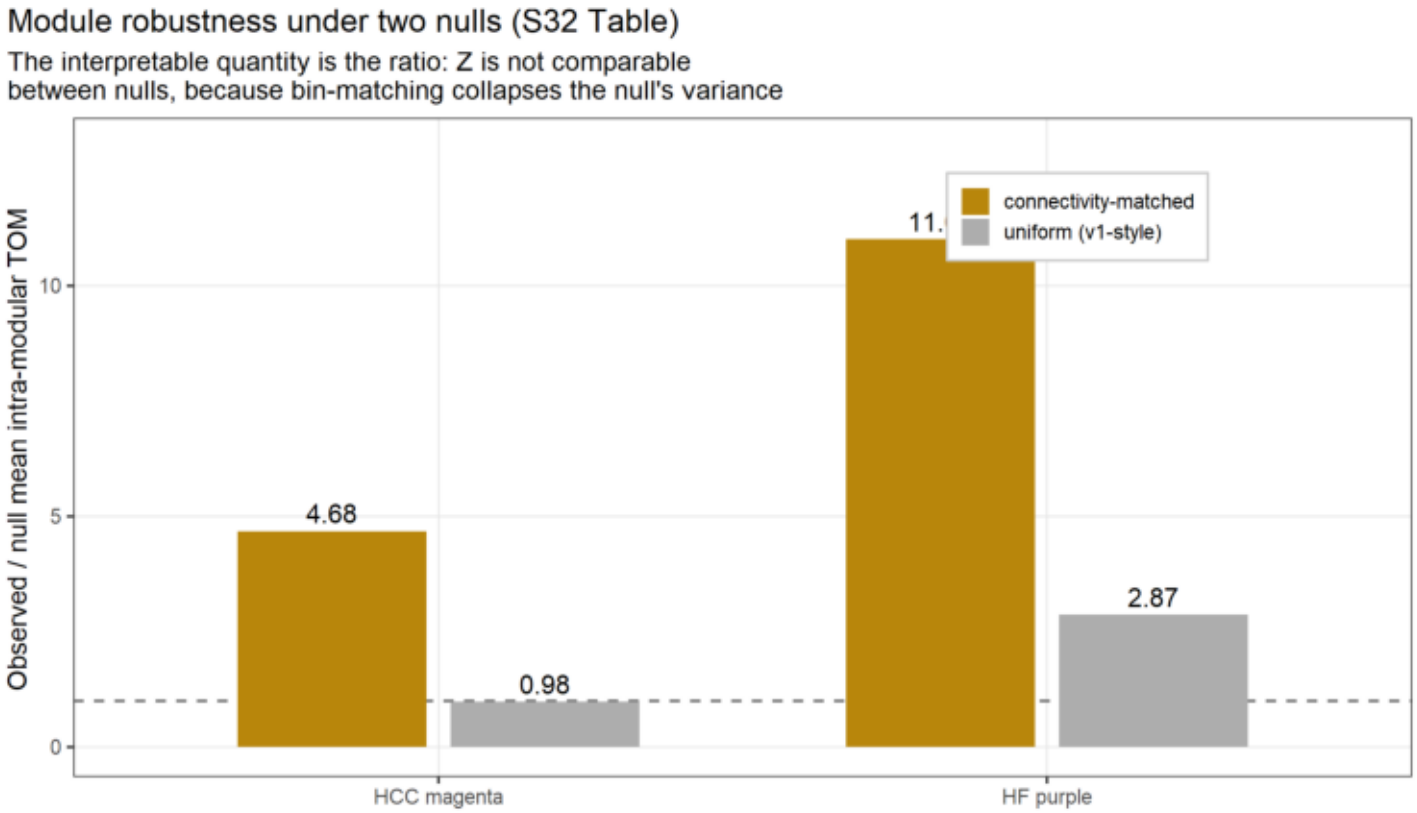
S3 Fig.

Soft-threshold selection. Scale-free topology fit and mean connectivity versus candidate power for the HF network (top) and the HCC network (bottom), both built on disease samples only on the common universe U. The rule-implied power is marked for each disease and the common power β = 14 used for both networks is marked in both panels.



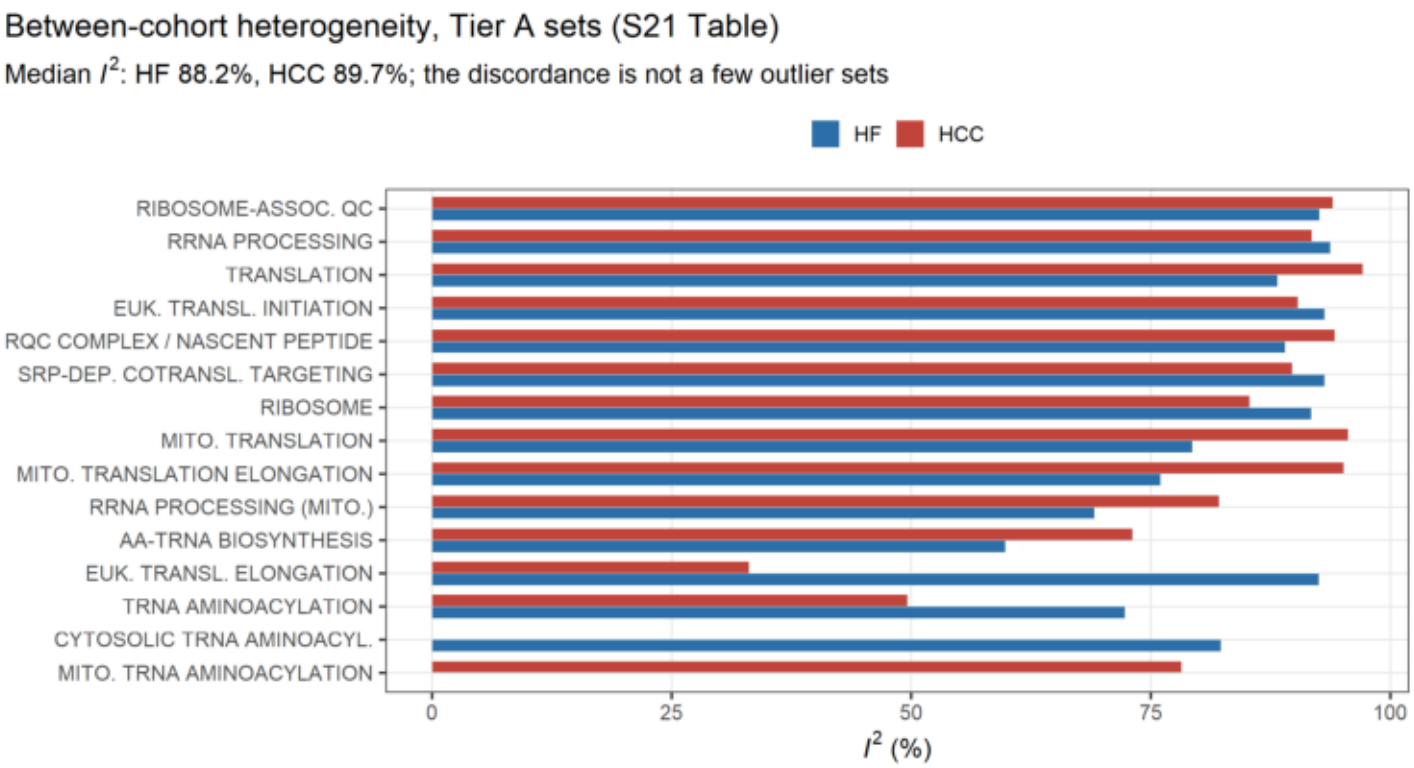
S4 Fig.

Module-internal robustness of the designated translation modules. Permutation distributions of the mean intra-modular topological overlap under the connectivity-matched null (primary) and the uniform null (secondary), for HF purple and HCC magenta; jackknife distributions of mean intra-modular connectivity.



S5 Fig.

Between-cohort heterogeneity. I^2 and Cochran's Q P-values for the 15 Tier A sets within each disease, with pooled effect sizes and confidence intervals.



S1 Table

Gene universe loss chain. Every filtering stage from platform feature to inclusion in the common universe U, separately for GPL11532 (HF array) and GSE141198 (HCC RNA-seq), reporting the number of features surviving at each step. The configuration hash for the frozen settings used throughout is recorded in the header.

Complete table: Table_S1_gene_universe_loss_chain.csv (10 rows x 3 columns).

platform	stage	n
GPL11532 (HF array)	features on platform (matrix rows)	33297
GPL11532 (HF array)	detected by expression filter	16632
GPL11532 (HF array)	mapped to a gene symbol	13229
GPL11532 (HF array)	unique symbols after collapsing	12090
GPL11532 (HF array)	present in common universe U	10213
GSE141198 (HCC RNA-seq)	features on platform (symbol-...	15036
GSE141198 (HCC RNA-seq)	detected by expression filter	13178
GSE141198 (HCC RNA-seq)	mapped to a gene symbol	13178
GSE141198 (HCC RNA-seq)	unique symbols after collapsing	13178
GSE141198 (HCC RNA-seq)	present in common universe U	10213

S2 Table (page 1 of 2)

A priori gene set definitions and coverage. Every scored gene set with its tier assignment (A, B, or C) and its measured gene count in each platform and in the common universe.

tier	set	n.genes.in msigdb	n.in.HF platform	n.in.HCC platform	n.in.common.U	frac.in.U
TierA	KEGG_RIBOSOME	88	81	85	80	0.909
TierA	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	41	40	41	40	0.976
TierA	REACTOME_TRANSLATION	414	385	406	382	0.923
TierA	...	140	132	136	130	0.929
TierA	...	115	108	110	105	0.913
TierA	REACTOME_TRNA_AMINOACYLATION	50	49	50	49	0.98
TierA	...	30	29	30	29	0.967
TierA	REACTOME_RRNA_PROCESSING	248	226	240	223	0.899
TierA	...	133	126	129	124	0.932
TierA	...	167	160	163	158	0.946
TierA	...	108	103	105	101	0.935
TierA	REACTOME_MITOCHONDRIAL_TRANSLATION	124	104	121	104	0.839
TierA	...	99	92	96	92	0.929
TierA	...	23	23	23	23	1
TierA	...	20	18	18	18	0.9
TierB	KEGG_RIBOSOME	88	81	85	80	0.909
TierB	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	41	40	41	40	0.976
TierB	REACTOME_TRANSLATION	414	385	406	382	0.923
TierB	...	140	132	136	130	0.929
TierB	...	115	108	110	105	0.913
TierB	REACTOME_TRNA_AMINOACYLATION	50	49	50	49	0.98
TierB	...	30	29	30	29	0.967

S2 Table (page 2 of 2)

A priori gene set definitions and coverage. Every scored gene set with its tier assignment (A, B, or C) and its measured gene count in each platform and in the common universe.

Printed from Table_S2_canonical_set_coverage.csv (69 rows x 7 columns): 44 of 69 rows. The CSV is the complete and authoritative version.

tier	set	n.genes.in msigdb	n.in.HF platform	n.in.HCC platform	n.in.common.U	frac.in.U
TierB	REACTOME_RRNA_PROCESSING	248	226	240	223	0.899
TierB	...	133	126	129	124	0.932
TierB	...	167	160	163	158	0.946
TierB	...	108	103	105	101	0.935
TierB	REACTOME_MITOCHONDRIAL_TRANSLATION	124	104	121	104	0.839
TierB	...	99	92	96	92	0.929
TierB	...	23	23	23	23	1
TierB	...	20	18	18	18	0.9
TierB	REACTOME_TRNA_PROCESSING	136	111	131	110	0.809
TierB	...	76	63	75	62	0.816
TierB	...	49	39	47	39	0.796
TierB	...	80	78	78	77	0.963
TierB	...	11	11	11	11	1
TierB	...	7	5	5	5	0.714
TierC_v1	...	47	27	25	20	0.426
TierC_v1	HALLMARK_MYC_TARGETS_V1	200	188	199	187	0.935
TierC_v1	...	30	29	30	29	0.967
TierC_v1	REACTOME_TRNA_AMINOACYLATION	50	49	50	49	0.98
TierC_v1	...	108	103	105	101	0.935
TierC_v1	...	167	160	163	158	0.946
TierC_v1	...	103	99	99	96	0.932
TierC_v1	...	95	84	90	83	0.874

S3 Table (page 1 of 2)

Tier A gene loss. Members of the 15 Tier A sets that are absent from the common universe U, listed per set with the reason for exclusion, so that the effect of the universe definition on the translation sets is auditable.

gene	in.HF platform	in.HCC platform	in.common.U	loss.reason
FAU	True	True	True	retained
MRPL13	False	True	False	lost on HF array (GPL11532)
RPL10	True	True	True	retained
RPL10A	True	True	True	retained
RPL10L	False	False	False	absent from both platforms
RPL11	True	True	True	retained
RPL12	True	True	True	retained
RPL13	True	True	True	retained
RPL13A	True	True	True	retained
RPL14	True	True	True	retained
RPL15	True	True	True	retained
RPL17	True	True	True	retained
RPL18	True	True	True	retained
RPL18A	True	True	True	retained
RPL19	True	True	True	retained
RPL21	True	True	True	retained
RPL22	False	True	False	lost on HF array (GPL11532)
RPL22L1	False	True	False	lost on HF array (GPL11532)
RPL23	True	True	True	retained
RPL23A	True	True	True	retained
RPL24	True	True	True	retained
RPL26	True	True	True	retained

S3 Table (page 2 of 2)

Tier A gene loss. Members of the 15 Tier A sets that are absent from the common universe U, listed per set with the reason for exclusion, so that the effect of the universe definition on the translation sets is auditable.

Printed from Table_S3_tierA_gene_loss.csv (497 rows x 5 columns): 44 of 497 rows. The CSV is the complete and authoritative version.

gene	in.HF platform	in.HCC platform	in.common.U	loss.reason
RPL26L1	True	True	True	retained
RPL27	True	True	True	retained
RPL27A	True	True	True	retained
RPL28	True	True	True	retained
RPL29	True	True	True	retained
RPL3	True	True	True	retained
RPL30	True	True	True	retained
RPL31	True	True	True	retained
RPL32	True	True	True	retained
RPL34	False	True	False	lost on HF array (GPL11532)
RPL35	True	True	True	retained
RPL35A	True	True	True	retained
RPL36	True	True	True	retained
RPL36A	True	True	True	retained
RPL36AL	True	True	True	retained
RPL37	True	True	True	retained
RPL37A	True	True	True	retained
RPL38	True	True	True	retained
RPL39	True	True	True	retained
RPL3L	True	False	False	lost in HCC RNA-seq
RPL4	True	True	True	retained
RPL41	True	True	True	retained

S4 Table (page 1 of 2)

Module inventory. Every module detected in each harmonised network, with its gene count, percentage of the universe, and the shared soft-threshold power.

disease	module	n.genes	power	pct.of.genes
HF	grey	2210	14	21.6
HF	turquoise	2132	14	20.9
HF	blue	1525	14	14.9
HF	brown	1516	14	14.8
HF	yellow	496	14	4.9
HF	green	425	14	4.2
HF	red	189	14	1.9
HF	black	184	14	1.8
HF	pink	173	14	1.7
HF	magenta	167	14	1.6
HF	purple	153	14	1.5
HF	greenyellow	150	14	1.5
HF	tan	146	14	1.4
HF	salmon	105	14	1
HF	cyan	103	14	1
HF	midnightblue	93	14	0.9
HF	lightcyan	73	14	0.7
HF	grey60	70	14	0.7
HF	lightgreen	70	14	0.7
HF	lightyellow	58	14	0.6
HF	royalblue	52	14	0.5
HF	darkred	43	14	0.4

S4 Table (page 2 of 2)

Module inventory. Every module detected in each harmonised network, with its gene count, percentage of the universe, and the shared soft-threshold power.

Printed from Table_S4_module_inventory.csv (45 rows x 5 columns): 44 of 45 rows. The CSV is the complete and authoritative version.

disease	module	n.genes	power	pct.of.genes
HF	darkgreen	41	14	0.4
HF	darkturquoise	39	14	0.4
HCC	turquoise	3771	14	36.9
HCC	blue	1948	14	19.1
HCC	grey	1445	14	14.1
HCC	brown	389	14	3.8
HCC	yellow	347	14	3.4
HCC	green	323	14	3.2
HCC	red	302	14	3
HCC	black	255	14	2.5
HCC	pink	254	14	2.5
HCC	magenta	146	14	1.4
HCC	purple	145	14	1.4
HCC	greenyellow	130	14	1.3
HCC	tan	117	14	1.1
HCC	salmon	110	14	1.1
HCC	cyan	99	14	1
HCC	midnightblue	96	14	0.9
HCC	lightcyan	90	14	0.9
HCC	grey60	78	14	0.8
HCC	lightgreen	68	14	0.7
HCC	lightyellow	57	14	0.6

S5 Table (page 1 of 2)

Soft-threshold fit. Signed scale-free topology fit at each candidate power for both diseases, with chosen_own, the power that disease's own soft-threshold rule selects, and chosen_common, the power both networks are actually built at. The rule takes the lowest power reaching the R^2 ≥ 0.85 cut and falls back to the best-fitting power when no candidate does, so chosen_own flags the rule's pick rather than the criterion: no HCC candidate reaches the cut, and its pick is the fallback (β = 5, R^2 = 0.790), while the common power is the stricter heart-failure pick (β = 14).

disease	power	fit	chosen.own	chosen.common
HF	1	-0.3254	False	False
HF	2	-0.1333	False	False
HF	3	-0.009481	False	False
HF	4	0.06626	False	False
HF	5	0.3137	False	False
HF	6	0.4806	False	False
HF	7	0.6016	False	False
HF	8	0.7002	False	False
HF	9	0.7548	False	False
HF	10	0.7983	False	False
HF	12	0.8443	False	False
HF	14	0.8607	True	True
HF	16	0.8774	False	False
HF	18	0.8852	False	False
HF	20	0.8828	False	False
HCC	1	-0.8678	False	False
HCC	2	0.5622	False	False
HCC	3	0.736	False	False
HCC	4	0.7813	False	False
HCC	5	0.7898	True	False
HCC	6	0.7618	False	False
HCC	7	0.7606	False	False

S5 Table (page 2 of 2)

Soft-threshold fit. Signed scale-free topology fit at each candidate power for both diseases, with chosen_own, the power that disease's own soft-threshold rule selects, and chosen_common, the power both networks are actually built at. The rule takes the lowest power reaching the R^2 ≥ 0.85 cut and falls back to the best-fitting power when no candidate does, so chosen_own flags the rule's pick rather than the criterion: no HCC candidate reaches the cut, and its pick is the fallback (β = 5, R^2 = 0.790), while the common power is the stricter heart-failure pick (β = 14).

Complete table: Table_S5_soft_threshold_fit.csv (30 rows x 5 columns).

disease	power	fit	chosen.own	chosen.common
HCC	8	0.755	False	False
HCC	9	0.7539	False	False
HCC	10	0.7555	False	False
HCC	12	0.7777	False	False
HCC	14	0.7783	False	True
HCC	16	0.7728	False	False
HCC	18	0.7784	False	False
HCC	20	0.7875	False	False

S6 Table (page 1 of 2)

Module–clinical-trait relationships in the HF network. Correlation of every HF module eigengene with ischaemic aetiology, sex, and age across the 177 failing ventricles. Disease status itself is not a trait here: the network was built on disease samples only, so all samples are failing myocardium and there is no within-network control group to correlate against. No module survives FDR correction against any of the three traits.

module	trait	r	p	fdr
MEpurple	ICM	-0.1002	0.1845	0.641
MElightgreen	ICM	0.1765	0.01876	0.5953
MEtan	ICM	0.2033	0.006651	0.4589
MEgreen	ICM	0.09785	0.1951	0.641
MEblack	ICM	0.1659	0.02729	0.5953
MEmidnightblue	ICM	0.02303	0.7609	0.8669
MEroyalblue	ICM	0.06373	0.3994	0.8266
MEbrown	ICM	0.06817	0.3673	0.792
MEmagenta	ICM	0.1081	0.1523	0.641
MEdarkgreen	ICM	-0.1003	0.1843	0.641
MEred	ICM	-0.1485	0.04854	0.5953
MEsalmon	ICM	-0.07541	0.3185	0.7325
MEblue	ICM	-0.03795	0.616	0.8571
MElightcyan	ICM	0.1316	0.08071	0.5953
MEpink	ICM	0.1198	0.1121	0.5953
MEturquoise	ICM	0.03653	0.6293	0.8571
MEcyan	ICM	0.0315	0.6773	0.8571
MEdarkturquoise	ICM	-0.08175	0.2794	0.7185
MEgrey60	ICM	-0.02961	0.6956	0.8571
MEyellow	ICM	-0.03363	0.6568	0.8571
MEdarkred	ICM	-0.0297	0.6947	0.8571
MEgreenyellow	ICM	-0.1202	0.1111	0.5953

S6 Table (page 2 of 2)

Module–clinical-trait relationships in the HF network. Correlation of every HF module eigengene with ischaemic aetiology, sex, and age across the 177 failing ventricles. Disease status itself is not a trait here: the network was built on disease samples only, so all samples are failing myocardium and there is no within-network control group to correlate against. No module survives FDR correction against any of the three traits.

Printed from Table_S6_module_trait_HF.csv (69 rows x 5 columns): 44 of 69 rows. The CSV is the complete and authoritative version.

module	trait	r	p	fdr
MElightyellow	ICM	-0.01234	0.8705	0.8964
MEpurple	male	-0.06267	0.4073	0.8266
MElightgreen	male	-0.1085	0.1505	0.641
MEtan	male	0.0271	0.7203	0.8575
MEgreen	male	-0.02705	0.7208	0.8575
MEblack	male	-0.08714	0.2488	0.7152
MEmidnightblue	male	0.01433	0.8499	0.8896
MEroyalblue	male	-0.05149	0.4961	0.8571
MEbrown	male	-0.04969	0.5113	0.8571
MEmagenta	male	0.03608	0.6335	0.8571
MEdarkgreen	male	-0.04476	0.5542	0.8571
MEred	male	0.04017	0.5955	0.8571
MEsalmon	male	0.0002034	0.9979	0.9979
MEblue	male	0.02463	0.7449	0.8669
MElightcyan	male	0.04728	0.532	0.8571
MEpink	male	-0.01961	0.7956	0.8735
MEturquoise	male	0.01596	0.833	0.8896
MEcyan	male	0.02249	0.7664	0.8669
MEdarkturquoise	male	0.04498	0.5522	0.8571
MEgrey60	male	0.04235	0.5757	0.8571
MEyellow	male	0.05305	0.4831	0.8571
MEdarkred	male	0.03873	0.6088	0.8571

S7 Table (page 1 of 2)

Gene module membership. Module assignment for every gene in the universe in each network, with intramodular connectivity and module membership.

disease	module	gene	kME
HF	black	ATP6AP2	0.7941
HF	black	TXNDC12	0.7661
HF	black	SSR1	0.7545
HF	black	MOB1A	0.7531
HF	black	STX12	0.7488
HF	black	COPB1	0.7433
HF	black	SNX2	0.7417
HF	black	CD164	0.7417
HF	black	SELENOF	0.7386
HF	black	CCNC	0.7372
HF	black	SGK3	0.7363
HF	black	TMX1	0.7332
HF	black	MDM2	0.7265
HF	black	TBK1	0.7252
HF	black	TMEM87A	0.7127
HF	black	ERO1A	0.7101
HF	black	NIP7	0.7063
HF	black	BZW1	0.6987
HF	black	NIFK	0.6908
HF	black	RAB2B	0.6902
HF	black	OAT	0.688
HF	black	ARF4	0.6842

S7 Table (page 2 of 2)

Gene module membership. Module assignment for every gene in the universe in each network, with intramodular connectivity and module membership.

Printed from Table_S7_gene_module_membership.csv (20426 rows x 4 columns): 44 of 20426 rows. The CSV is the complete and authoritative version.

disease	module	gene	kME
HF	black	ATF6	0.6833
HF	black	HMGN4	0.6793
HF	black	ERP44	0.6772
HF	black	WDR75	0.6769
HF	black	HACD3	0.6763
HF	black	UTP18	0.6745
HF	black	ACTR3	0.6744
HF	black	GNAI3	0.671
HF	black	SRFBP1	0.6689
HF	black	MTDH	0.6644
HF	black	PGM3	0.6636
HF	black	LMAN2	0.6634
HF	black	GABPB1	0.6619
HF	black	DCK	0.655
HF	black	CYRIB	0.6542
HF	black	ARFGAP3	0.6493
HF	black	UBLCP1	0.6488
HF	black	VAMP4	0.647
HF	black	SNX6	0.6463
HF	black	RPL7L1	0.6449
HF	black	NOC3L	0.6412
HF	black	UFM1	0.6392

S8 Table (page 1 of 2)

Module x gene-set correlation matrix. The correlation between each module eigengene and the ssGSEA score of each a priori translation set, with a sample-permutation P value, for both networks; this was the secondary (activity-based) designation criterion. The primary criterion was the hypergeometric over-representation test in S9 Table, from which the translation modules were designated without manual input.

set	module	r	tier	disease	p.fwer	module.agg	p.module
KEGG_RIBOSOME	purple	0.9294	TierA	HF	0.000999	0.5034	0.000999
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	purple	-0.2787	TierA	HF	0.03197	0.5034	0.000999
REACTOME_TRANSLATION	purple	0.2061	TierA	HF	0.4356	0.5034	0.000999
...	purple	0.9283	TierA	HF	0.000999	0.5034	0.000999
...	purple	0.9207	TierA	HF	0.000999	0.5034	0.000999
REACTOME_TRNA_AMINOACYLATION	purple	-0.2221	TierA	HF	0.2637	0.5034	0.000999
...	purple	-0.02213	TierA	HF	1	0.5034	0.000999
REACTOME_RRNA_PROCESSING	purple	0.729	TierA	HF	0.000999	0.5034	0.000999
...	purple	0.9465	TierA	HF	0.000999	0.5034	0.000999
...	purple	0.8207	TierA	HF	0.000999	0.5034	0.000999
...	purple	0.6753	TierA	HF	0.000999	0.5034	0.000999
REACTOME_MITOCHONDRIAL_TRANSLATION	purple	-0.1229	TierA	HF	0.995	0.5034	0.000999
...	purple	-0.08344	TierA	HF	1	0.5034	0.000999
...	purple	-0.3336	TierA	HF	0.003996	0.5034	0.000999
...	purple	-0.3322	TierA	HF	0.003996	0.5034	0.000999
KEGG_RIBOSOME	lightgreen	0.2565	TierA	HF	0.08192	0.3275	0.000999
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	lightgreen	-0.5188	TierA	HF	0.000999	0.3275	0.000999
REACTOME_TRANSLATION	lightgreen	-0.3996	TierA	HF	0.000999	0.3275	0.000999
...	lightgreen	0.2081	TierA	HF	0.4036	0.3275	0.000999
...	lightgreen	0.2347	TierA	HF	0.1728	0.3275	0.000999
REACTOME_TRNA_AMINOACYLATION	lightgreen	-0.48	TierA	HF	0.000999	0.3275	0.000999
...	lightgreen	-0.2445	TierA	HF	0.1229	0.3275	0.000999

S8 Table (page 2 of 2)

Module x gene-set correlation matrix. The correlation between each module eigengene and the ssGSEA score of each a priori translation set, with a sample-permutation P value, for both networks; this was the secondary (activity-based) designation criterion. The primary criterion was the hypergeometric over-representation test in S9 Table, from which the translation modules were designated without manual input.

Printed from Table_S8_set_module_correlation.csv (2967 rows x 8 columns): 44 of 2967 rows. The CSV is the complete and authoritative version.

set	module	r	tier	disease	p.fwer	module.agg	p.module
REACTOME_RRNA_PROCESSING	lightgreen	0.02247	TierA	HF	1	0.3275	0.000999
...	lightgreen	0.2685	TierA	HF	0.04895	0.3275	0.000999
...	lightgreen	0.01078	TierA	HF	1	0.3275	0.000999
...	lightgreen	-0.08884	TierA	HF	1	0.3275	0.000999
REACTOME_MITOCHONDRIAL_TRANSLATION	lightgreen	-0.511	TierA	HF	0.000999	0.3275	0.000999
...	lightgreen	-0.4983	TierA	HF	0.000999	0.3275	0.000999
...	lightgreen	-0.5887	TierA	HF	0.000999	0.3275	0.000999
...	lightgreen	-0.581	TierA	HF	0.000999	0.3275	0.000999
KEGG_RIBOSOME	tan	0.1877	TierA	HF	0.6454	0.302	0.000999
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	tan	-0.468	TierA	HF	0.000999	0.302	0.000999
REACTOME_TRANSLATION	tan	-0.4171	TierA	HF	0.000999	0.302	0.000999
...	tan	0.151	TierA	HF	0.9361	0.302	0.000999
...	tan	0.1665	TierA	HF	0.8432	0.302	0.000999
REACTOME_TRNA_AMINOACYLATION	tan	-0.465	TierA	HF	0.000999	0.302	0.000999
...	tan	-0.3454	TierA	HF	0.002997	0.302	0.000999
REACTOME_RRNA_PROCESSING	tan	-0.023	TierA	HF	1	0.302	0.000999
...	tan	0.154	TierA	HF	0.9271	0.302	0.000999
...	tan	-0.09133	TierA	HF	1	0.302	0.000999
...	tan	-0.1824	TierA	HF	0.7003	0.302	0.000999
REACTOME_MITOCHONDRIAL_TRANSLATION	tan	-0.4723	TierA	HF	0.000999	0.302	0.000999
...	tan	-0.4649	TierA	HF	0.000999	0.302	0.000999
...	tan	-0.4812	TierA	HF	0.000999	0.302	0.000999

S9 Table (page 1 of 2)

Translation over-representation by module. One-sided hypergeometric test of Tier A set membership within each module, with fold enrichment and BH-corrected P-values, for both networks.

module	n.module	n.overlap	expected	fold	p	fdr	disease
purple	153	73	6.8	10.69	2.367e-58	5.682e-57	HF
blue	1525	115	68.1	1.69	2.921e-09	3.505e-08	HF
pink	173	23	7.7	2.98	2.512e-06	2.01e-05	HF
turquoise	2132	119	95.2	1.25	0.003604	0.02162	HF
black	184	14	8.2	1.7	0.03598	0.1727	HF
midnightblue	93	7	4.2	1.69	0.1214	0.4856	HF
lightyellow	58	3	2.6	1.16	0.4828	1	HF
lightcyan	73	2	3.3	0.61	0.8438	1	HF
greenyellow	150	4	6.7	0.6	0.9078	1	HF
salmon	105	2	4.7	0.43	0.952	1	HF
cyan	103	1	4.6	0.22	0.9912	1	HF
yellow	496	7	22.1	0.32	1	1	HF
green	425	5	19	0.26	1	1	HF
grey	2210	65	98.7	0.66	1	1	HF
brown	1516	16	67.7	0.24	1	1	HF
red	189	0	8.4	0	1	1	HF
magenta	167	0	7.5	0	1	1	HF
tan	146	0	6.5	0	1	1	HF
grey60	70	0	3.1	0	1	1	HF
lightgreen	70	0	3.1	0	1	1	HF
royalblue	52	0	2.3	0	1	1	HF
darkred	43	0	1.9	0	1	1	HF

S9 Table (page 2 of 2)

Translation over-representation by module. One-sided hypergeometric test of Tier A set membership within each module, with fold enrichment and BH-corrected P-values, for both networks.

Printed from Table_S9_translation_overrepresentation.csv (45 rows x 8 columns): 44 of 45 rows. The CSV is the complete and authoritative version.

module	n.module	n.overlap	expected	fold	p	fdr	disease
darkgreen	41	0	1.8	0	1	1	HF
darkturquoise	39	0	1.7	0	1	1	HF
blue	1948	173	87	1.99	4.698e-22	9.866e-21	HCC
magenta	146	26	6.5	3.99	1.165e-09	1.223e-08	HCC
brown	389	37	17.4	2.13	1.053e-05	7.368e-05	HCC
lightgreen	68	8	3	2.63	0.01047	0.05496	HCC
grey60	78	8	3.5	2.3	0.02262	0.09502	HCC
cyan	99	8	4.4	1.81	0.07454	0.2609	HCC
salmon	110	6	4.9	1.22	0.368	1	HCC
lightyellow	57	3	2.5	1.18	0.4712	1	HCC
royalblue	43	2	1.9	1.04	0.5783	1	HCC
pink	254	8	11.3	0.71	0.8861	1	HCC
green	323	9	14.4	0.62	0.9558	1	HCC
purple	145	2	6.5	0.31	0.99	1	HCC
black	255	5	11.4	0.44	0.9903	1	HCC
grey	1445	44	64.5	0.68	0.9987	1	HCC
yellow	347	1	15.5	0.06	1	1	HCC
turquoise	3771	116	168.4	0.69	1	1	HCC
red	302	0	13.5	0	1	1	HCC
greenyellow	130	0	5.8	0	1	1	HCC
tan	117	0	5.2	0	1	1	HCC
midnightblue	96	0	4.3	0	1	1	HCC

S10 Table (page 1 of 2)

Translation module gene lists. The genes of HF purple (153) and HCC magenta (146), with their connectivity and membership statistics in their own and the other network.

disease	module	gene
HF	purple	AHCY
HF	purple	AIP
HF	purple	ASL
HF	purple	ATOX1
HF	purple	BAD
HF	purple	BLOC1S1
HF	purple	BLOC1S3
HF	purple	BLVRA
HF	purple	BTF3
HF	purple	C12orf57
HF	purple	CCT2
HF	purple	CCT3
HF	purple	CNIH1
HF	purple	COA3
HF	purple	COMMD6
HF	purple	COPS9
HF	purple	CWC15
HF	purple	DDX49
HF	purple	DPP7
HF	purple	DUSP23
HF	purple	DYNLT1
HF	purple	EEF1B2

S10 Table (page 2 of 2)

Translation module gene lists. The genes of HF purple (153) and HCC magenta (146), with their connectivity and membership statistics in their own and the other network.

Printed from Table_S10_translation_module_genes.csv (299 rows x 3 columns): 44 of 299 rows. The CSV is the complete and authoritative version.

disease	module	gene
HF	purple	EEF1D
HF	purple	EIF3D
HF	purple	EIF3I
HF	purple	EIF4EBP1
HF	purple	ELOB
HF	purple	EVA1B
HF	purple	FAM216A
HF	purple	FKBP8
HF	purple	GET3
HF	purple	GPBP1
HF	purple	GRAMD1A
HF	purple	GUK1
HF	purple	HBB
HF	purple	HCFC1R1
HF	purple	HINT1
HF	purple	HSBP1
HF	purple	IMPDH2
HF	purple	JTB
HF	purple	LAMTOR1
HF	purple	LGALS1
HF	purple	LSM10
HF	purple	MAGEH1

S11 Table

Cohort coverage. Every cohort with its sample count, group sizes, technology, and platform, including the platforms GSE14520 is split into.

Complete table: Table_S11_cohort_coverage.csv (8 rows x 11 columns).

cohort	disease	tech	n.genes	n.samples	n.disease	n.control	n.unlabeled	tierA.covered	tierA.total	tierA.pct
GSE57338	HF	array	12090	313	177	136	0	459	497	92.4
GSE141198	HCC	RNA-seq	13178	148	148	0	0	487	497	98
TCGA_LIHC	HCC	RNA-seq	17146	421	371	50	0	483	497	97.2
GSE116250	HF	RNA-seq	26261	64	50	14	0	439	497	88.3
GSE14520_GPL3921	HCC	array	7629	445	225	220	0	299	497	60.2
GSE14520_GPL571	HCC	array	7733	41	22	19	0	322	497	64.8
GSE76427	HCC	array	16907	167	115	52	0	417	497	83.9
GSE141910	HF	RNA-seq	16978	366	200	166	0	482	497	97

S12 Table (page 1 of 2)

Set coverage per cohort. The number of measured genes for each scored set in each cohort, so that a set's contribution to any cohort-level statistic is auditable.

cohort	set	n.genes.in cohort	scored
GSE57338	KEGG_RIBOSOME	81	True
GSE57338	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	40	True
GSE57338	REACTOME_TRANSLATION	342	True
GSE57338	...	112	True
GSE57338	...	87	True
GSE57338	REACTOME_TRNA_AMINOACYLATION	41	True
GSE57338	...	23	True
GSE57338	REACTOME_RRNA_PROCESSING	196	True
GSE57338	...	106	True
GSE57338	...	138	True
GSE57338	...	98	True
GSE57338	REACTOME_MITOCHONDRIAL_TRANSLATION	95	True
GSE57338	...	83	True
GSE57338	...	21	True
GSE57338	...	17	True
GSE57338	REACTOME_TRNA_PROCESSING	97	True
GSE57338	...	52	True
GSE57338	...	36	True
GSE57338	...	58	True
GSE57338	...	11	True
GSE57338	...	5	True
GSE57338	...	27	True

S12 Table (page 2 of 2)

Set coverage per cohort. The number of measured genes for each scored set in each cohort, so that a set's contribution to any cohort-level statistic is auditable.

Printed from Table_S12_set_coverage_per_cohort.csv (328 rows x 4 columns): 44 of 328 rows. The CSV is the complete and authoritative version.

cohort	set	n.genes.in cohort	scored
GSE57338	HALLMARK_MYC_TARGETS_V1	188	True
GSE57338	...	82	True
GSE57338	...	76	True
GSE57338	...	17	True
GSE57338	HALLMARK_MYC_TARGETS_V2	53	True
GSE57338	...	44	True
GSE57338	...	30	True
GSE57338	...	58	True
GSE57338	...	35	True
GSE57338	HALLMARK_MTORC1_SIGNALING	180	True
GSE57338	...	13	True
GSE57338	...	9	True
GSE57338	...	43	True
GSE57338	...	43	True
GSE57338	...	9	True
GSE57338	...	29	True
GSE57338	...	9	True
GSE57338	...	8	True
GSE57338	...	9	True
GSE141198	KEGG_RIBOSOME	85	True
GSE141198	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	41	True
GSE141198	REACTOME_TRANSLATION	364	True

S13 Table

Scored set counts. The number of sets scored per cohort, broken down by tier, with any set dropped for insufficient measured genes.

Complete table: Table_S13_scored_set_counts.csv (8 rows x 5 columns).

cohort	n.sets.total	n.tierA	n.tierB	n.tierC
GSE57338	41	15	21	33
GSE141198	41	15	21	33
TCGA_LIHC	41	15	21	33
GSE116250	40	15	20	33
GSE14520_GPL3921	34	14	19	27
GSE14520_GPL571	36	15	20	29
GSE76427	35	14	19	28
GSE141910	41	15	21	33

S14 Table (columns 1 of 2)

Within-disease effect sizes. Hedges g with n-weighted pooled SD, the small-sample correction J, bootstrap confidence intervals, and test statistics for every set in every cohort.

Printed from Table_S14_effect_sizes_within_disease.csv (268 rows x 23 columns): 22 of 268 rows. The CSV is the complete and authoritative version.

cohort	disease	tech	set	tier	n.disease	n.control	mean.disease	mean.control	sd.disease	sd.control	cohens.d equalweight
GSE57338	HF	array	KEGG_RIBOSOME	A	177	136	0.5646	0.5741	0.0186	0.02437	-0.4394
GSE57338	HF	array	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	177	136	0.2202	0.2253	0.02426	0.02403	-0.2086
GSE57338	HF	array	REACTOME_TRANSLATION	A	177	136	0.4068	0.4156	0.01644	0.01585	-0.5403
GSE57338	HF	array	...	A	177	136	0.5612	0.5725	0.01395	0.02099	-0.6342
GSE57338	HF	array	...	A	177	136	0.5774	0.5867	0.01697	0.02238	-0.4672
GSE57338	HF	array	REACTOME_TRNA_AMINOACYLATION	A	177	136	0.2053	0.2137	0.02271	0.0243	-0.3604
GSE57338	HF	array	...	A	177	136	0.3976	0.4241	0.01923	0.03777	-0.8835
GSE57338	HF	array	REACTOME_RRNA_PROCESSING	A	177	136	0.2403	0.2588	0.012	0.03178	-0.7702
GSE57338	HF	array	...	A	177	136	0.5349	0.5467	0.01535	0.02382	-0.5904
GSE57338	HF	array	...	A	177	136	0.5226	0.5318	0.01232	0.0183	-0.5843
GSE57338	HF	array	...	A	177	136	0.5239	0.5318	0.01239	0.01635	-0.5421
GSE57338	HF	array	REACTOME_MITOCHONDRIAL_TRANSLATION	A	177	136	0.2456	0.2489	0.0396	0.03456	-0.08902
GSE57338	HF	array	...	A	177	136	0.2579	0.2609	0.03892	0.03476	-0.08181
GSE57338	HF	array	...	A	177	136	0.03087	0.01304	0.03422	0.02874	0.5645
GSE57338	HF	array	...	A	177	136	-0.05406	-0.04939	0.05149	0.03886	-0.1023
GSE57338	HF	array	REACTOME_TRNA_PROCESSING	B_extra	177	136	-0.0367	-0.02758	0.01723	0.02446	-0.4312
GSE57338	HF	array	...	B_extra	177	136	0.08264	0.09369	0.01417	0.02392	-0.5622
GSE57338	HF	array	...	B_extra	177	136	-0.2042	-0.2005	0.02095	0.03245	-0.1379
GSE57338	HF	array	...	B_extra	177	136	-0.03546	-0.0005126	0.01828	0.05765	-0.8172
GSE57338	HF	array	...	B_extra	177	136	-0.1646	-0.1413	0.05361	0.0449	-0.4705
GSE57338	HF	array	...	B_extra	177	136	0.04544	0.06177	0.04657	0.04029	-0.3749
GSE57338	HF	array	...	C_v1	177	136	0.07477	0.09175	0.02215	0.0305	-0.6368

S14 Table (columns 2 of 2)

Within-disease effect sizes. Hedges g with n-weighted pooled SD, the small-sample correction J, bootstrap confidence intervals, and test statistics for every set in every cohort.

Printed from Table_S14_effect_sizes_within_disease.csv (268 rows x 23 columns): 22 of 268 rows. The CSV is the complete and authoritative version.

cohort	pooled.sd nweighted	hedges.J	d.nweighted	hedges.g	ci.lo	ci.hi	p.ttest	ci.excludes.0	meets.min effect	mirror candidate	p.fdr
GSE57338	0.0213	0.9976	-0.4473	-0.4462	-0.6826	-0.2193	0.0001906	True	True	True	0.0002894
GSE57338	0.02416	0.9976	-0.2085	-0.208	-0.4249	0.02243	0.06822	False	True	False	0.08475
GSE57338	0.01619	0.9976	-0.539	-0.5377	-0.7463	-0.3254	3.18e-06	True	True	True	6.154e-06
GSE57338	0.01736	0.9976	-0.6511	-0.6495	-0.8878	-0.4249	1.499e-07	True	True	True	4.389e-07
GSE57338	0.0195	0.9976	-0.4757	-0.4746	-0.7172	-0.2489	7.56e-05	True	True	True	0.0001192
GSE57338	0.02342	0.9976	-0.3621	-0.3612	-0.5781	-0.138	0.001827	True	True	True	0.002496
GSE57338	0.02879	0.9976	-0.9199	-0.9177	-1.138	-0.7103	2.977e-12	True	True	True	1.526e-11
GSE57338	0.0228	0.9976	-0.8114	-0.8094	-1.007	-0.6242	1.223e-09	True	True	True	4.557e-09
GSE57338	0.01948	0.9976	-0.6071	-0.6057	-0.8485	-0.3791	9.641e-07	True	True	True	2.196e-06
GSE57338	0.01521	0.9976	-0.5993	-0.5979	-0.83	-0.3833	1.143e-06	True	True	True	2.467e-06
GSE57338	0.01424	0.9976	-0.5521	-0.5507	-0.7818	-0.3368	4.927e-06	True	True	True	8.783e-06
GSE57338	0.0375	0.9976	-0.08824	-0.08803	-0.2911	0.1401	0.4315	False	False	False	0.4655
GSE57338	0.03717	0.9976	-0.08121	-0.08101	-0.2864	0.1497	0.4704	False	False	False	0.4911
GSE57338	0.03196	0.9976	0.5582	0.5569	0.3469	0.7993	9.278e-07	True	True	True	2.196e-06
GSE57338	0.04643	0.9976	-0.1005	-0.1003	-0.3014	0.1268	0.3614	False	False	False	0.4005
GSE57338	0.02068	0.9976	-0.4411	-0.4401	-0.6467	-0.2139	0.0002682	True	True	True	0.0003927
GSE57338	0.01903	0.9976	-0.5808	-0.5794	-0.7853	-0.3714	3.302e-06	True	True	True	6.154e-06
GSE57338	0.02656	0.9976	-0.1418	-0.1414	-0.3563	0.1073	0.2402	False	False	False	0.2735
GSE57338	0.04039	0.9976	-0.8652	-0.8631	-1.05	-0.6899	1.977e-10	True	True	True	8.107e-10
GSE57338	0.05002	0.9976	-0.4652	-0.464	-0.6902	-0.2378	3.889e-05	True	True	True	6.378e-05
GSE57338	0.04396	0.9976	-0.3714	-0.3705	-0.591	-0.1429	0.00101	True	True	True	0.001428
GSE57338	0.02611	0.9976	-0.6502	-0.6486	-0.9084	-0.4232	1.117e-07	True	True	True	3.523e-07

S15 Table (page 1 of 2)

Set redundancy clusters. Jaccard similarity hierarchical clustering of the scored sets at cut height 0.7, giving the redundancy clusters used for the cluster-level mirror test.

set	tier	cluster	is representative	n.genes
REACTOME_RRNA_PROCESSING	A	1	True	248
...	A	1	False	167
...	A	1	False	140
...	A	1	False	133
...	A	1	False	115
...	A	1	False	108
KEGG_RIBOSOME	A	1	False	88
REACTOME_TRNA_AMINOACYLATION	A	2	True	50
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	2	False	41
...	A	2	False	30
...	A	2	False	23
REACTOME_TRANSLATION	A	3	True	414
REACTOME_MITOCHONDRIAL_TRANSLATION	A	4	True	124
...	A	4	False	99
...	A	5	True	20
...	B_extra	5	False	7
REACTOME_TRNA_PROCESSING	B_extra	6	True	136
...	B_extra	6	False	76
...	B_extra	7	True	49
...	B_extra	8	True	80
...	B_extra	9	True	11
...	C_v1	10	True	47

S15 Table (page 2 of 2)

Set redundancy clusters. Jaccard similarity hierarchical clustering of the scored sets at cut height 0.7, giving the redundancy clusters used for the cluster-level mirror test.

Complete table: Table_S15_set_redundancy_clusters.csv (41 rows x 5 columns).

set	tier	cluster	is representative	n.genes
HALLMARK_MYC_TARGETS_V1	C_v1	11	True	200
...	C_v1	12	True	103
...	C_v1	13	True	95
...	C_v1	14	True	23
HALLMARK_MYC_TARGETS_V2	C_v1	15	True	58
...	C_v1	16	True	71
...	C_v1	16	False	54
...	C_v1	17	True	40
...	C_v1	18	True	64
...	C_v1	18	False	30
HALLMARK_MTORC1_SIGNALING	C_v1	19	True	200
...	C_v1	20	True	15
...	C_v1	21	True	15
...	C_v1	21	False	14
...	C_v1	21	False	10
...	C_v1	21	False	10
...	C_v1	21	False	9
...	C_v1	22	True	108
...	C_v1	23	True	67

S16 Table

Effective number of tests. Nyholt and Li & Ji effective test counts per cohort and tier, alongside the nominal set counts.

Complete table: Table_S16_effective_tests.csv (8 rows x 7 columns).

cohort	tierA.m	tierA.nyholt	tierA.li.ji	all.m	all.nyholt	all.li.ji
GSE57338	15	10.25	5	41	32.18	13
GSE141198	15	10.54	6	41	33.47	15
TCGA_LIHC	15	8.63	5	41	32.56	17
GSE116250	15	10.37	6	40	35.18	17
GSE14520_GPL3921	14	9.3	6	34	25.43	15
GSE14520_GPL571	15	8.53	6	36	25.42	13
GSE76427	14	7.99	5	35	26.77	16
GSE141910	15	8.05	4	41	28.63	13

S17 Table (page 1 of 3)

Mirror sets. Per-set effect sizes in both diseases, the mirror flag, and the redundancy cluster assignment, for each of the four analysed pairings and every tier. The four are TCGA-LIHC x GSE57338 (the primary v1-cohort pairing), GSE76427 x GSE116250 and GSE14520_GPL3921 x GSE141910 (HCC array against an independent HF RNA-seq cohort), and GSE14520_GPL571 x GSE57338 (HCC array against the HF array cohort). Fig 2d gives the effect-size correlation and Tier A mirror count for all twelve HCC x HF combinations; the permutation tests and effective counts in S18 are computed on these four.

set	g.HCC	g.HF	mirror	tier	cluster	hcc.cohort	hf.cohort
KEGG_RIBOSOME	0.6393	-0.4462	True	A	1	TCGA_LIHC	GSE57338
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	1.024	-0.208	False	A	2	TCGA_LIHC	GSE57338
REACTOME_TRANSLATION	0.6862	-0.5377	True	A	3	TCGA_LIHC	GSE57338
...	0.7336	-0.6495	True	A	1	TCGA_LIHC	GSE57338
...	0.8797	-0.4746	True	A	1	TCGA_LIHC	GSE57338
REACTOME_TRNA_AMINOACYLATION	1.119	-0.3612	True	A	2	TCGA_LIHC	GSE57338
...	1.173	-0.9177	True	A	2	TCGA_LIHC	GSE57338
REACTOME_RRNA_PROCESSING	0.8342	-0.8094	True	A	1	TCGA_LIHC	GSE57338
...	0.8564	-0.6057	True	A	1	TCGA_LIHC	GSE57338
...	0.9669	-0.5979	True	A	1	TCGA_LIHC	GSE57338
...	1.1	-0.5507	True	A	1	TCGA_LIHC	GSE57338
REACTOME_MITOCHONDRIAL_TRANSLATION	0.1234	-0.08803	False	A	4	TCGA_LIHC	GSE57338
...	0.1764	-0.08101	False	A	4	TCGA_LIHC	GSE57338
...	0.6559	0.5569	False	A	2	TCGA_LIHC	GSE57338
...	-0.2577	-0.1003	False	A	5	TCGA_LIHC	GSE57338
KEGG_RIBOSOME	1.013	0.2236	False	A	1	GSE76427	GSE116250
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	1.042	0.2177	False	A	2	GSE76427	GSE116250
REACTOME_TRANSLATION	1.603	-0.4416	False	A	3	GSE76427	GSE116250
...	1.206	-0.04907	False	A	1	GSE76427	GSE116250
...	1.189	0.4075	False	A	1	GSE76427	GSE116250
REACTOME_TRNA_AMINOACYLATION	1.377	0.1044	False	A	2	GSE76427	GSE116250
REACTOME_RRNA_PROCESSING	1.265	-0.02748	False	A	1	GSE76427	GSE116250

S17 Table (page 2 of 3)

Mirror sets. Per-set effect sizes in both diseases, the mirror flag, and the redundancy cluster assignment, for each of the four analysed pairings and every tier. The four are TCGA-LIHC x GSE57338 (the primary v1-cohort pairing), GSE76427 x GSE116250 and GSE14520_GPL3921 x GSE141910 (HCC array against an independent HF RNA-seq cohort), and GSE14520_GPL571 x GSE57338 (HCC array against the HF array cohort). Fig 2d gives the effect-size correlation and Tier A mirror count for all twelve HCC x HF combinations; the permutation tests and effective counts in S18 are computed on these four.

set	g.HCC	g.HF	mirror	tier	cluster	hcc.cohort	hf.cohort
...	1.341	0.3627	False	A	1	GSE76427	GSE116250
...	1.412	0.3047	False	A	1	GSE76427	GSE116250
...	1.53	0.21	False	A	1	GSE76427	GSE116250
REACTOME_MITOCHONDRIAL_TRANSLATION	1.081	-0.8947	True	A	4	GSE76427	GSE116250
...	1.104	-0.7284	True	A	4	GSE76427	GSE116250
...	1.175	0.4217	False	A	2	GSE76427	GSE116250
...	-0.07637	-0.8351	False	A	5	GSE76427	GSE116250
KEGG_RIBOSOME	1.348	0.3286	False	A	1	GSE14520_GPL3921	GSE141910
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	0.5985	0.1062	False	A	2	GSE14520_GPL3921	GSE141910
REACTOME_TRANSLATION	2.34	0.06818	False	A	3	GSE14520_GPL3921	GSE141910
...	1.599	0.2034	False	A	1	GSE14520_GPL3921	GSE141910
...	1.07	0.3332	False	A	1	GSE14520_GPL3921	GSE141910
REACTOME_TRNA_AMINOACYLATION	1.392	0.0232	False	A	2	GSE14520_GPL3921	GSE141910
REACTOME_RRNA_PROCESSING	1.71	-0.01461	False	A	1	GSE14520_GPL3921	GSE141910
...	1.885	0.2271	False	A	1	GSE14520_GPL3921	GSE141910
...	2.355	0.184	False	A	1	GSE14520_GPL3921	GSE141910
...	2.499	0.08908	False	A	1	GSE14520_GPL3921	GSE141910
REACTOME_MITOCHONDRIAL_TRANSLATION	1.371	0.0652	False	A	4	GSE14520_GPL3921	GSE141910
...	1.355	0.1153	False	A	4	GSE14520_GPL3921	GSE141910
...	0.5985	0.5104	False	A	2	GSE14520_GPL3921	GSE141910
...	-0.5951	-0.2196	False	A	5	GSE14520_GPL3921	GSE141910
KEGG_RIBOSOME	0.6263	-0.4462	True	A	1	GSE14520_GPL571	GSE57338

S17 Table (page 3 of 3)

Mirror sets. Per-set effect sizes in both diseases, the mirror flag, and the redundancy cluster assignment, for each of the four analysed pairings and every tier. The four are TCGA-LIHC x GSE57338 (the primary v1-cohort pairing), GSE76427 x GSE116250 and GSE14520_GPL3921 x GSE141910 (HCC array against an independent HF RNA-seq cohort), and GSE14520_GPL571 x GSE57338 (HCC array against the HF array cohort). Fig 2d gives the effect-size correlation and Tier A mirror count for all twelve HCC x HF combinations; the permutation tests and effective counts in S18 are computed on these four.

Printed from Table_S17_mirror_sets.csv (175 rows x 11 columns): 8 of 11 columns and 58 of 175 rows. Rows cited in the manuscript are printed first. The CSV is the complete and authoritative version.

set	g.HCC	g.HF	mirror	tier	cluster	hcc.cohort	hf.cohort
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	0.6662	-0.208	False	A	2	GSE14520_GPL571	GSE57338
REACTOME_TRANSLATION	1.203	-0.5377	True	A	3	GSE14520_GPL571	GSE57338
...	0.9839	-0.6495	True	A	1	GSE14520_GPL571	GSE57338
...	0.444	-0.4746	False	A	1	GSE14520_GPL571	GSE57338
REACTOME_TRNA_AMINOACYLATION	0.6957	-0.3612	True	A	2	GSE14520_GPL571	GSE57338
...	1.29	-0.9177	True	A	2	GSE14520_GPL571	GSE57338
REACTOME_RRNA_PROCESSING	1.244	-0.8094	True	A	1	GSE14520_GPL571	GSE57338
...	0.9826	-0.6057	True	A	1	GSE14520_GPL571	GSE57338
...	1.105	-0.5979	True	A	1	GSE14520_GPL571	GSE57338
...	1.247	-0.5507	True	A	1	GSE14520_GPL571	GSE57338
REACTOME_MITOCHONDRIAL_TRANSLATION	0.8371	-0.08803	False	A	4	GSE14520_GPL571	GSE57338
...	0.8308	-0.08101	False	A	4	GSE14520_GPL571	GSE57338
...	0.05438	0.5569	False	A	2	GSE14520_GPL571	GSE57338
...	0.3713	-0.1003	False	A	5	GSE14520_GPL571	GSE57338

S18 Table

Mirror summary. For each of the four pairings named in S17: number of sets, clusters, mirror sets and mirror clusters, the sample-level permutation P-values at both levels, the effective test counts, and — reported for transparency only — the binomial P-value v1-style, which is not used for inference. The effective counts are computed per pairing on the measured sets of both cohorts, so they differ from the per-cohort values in S16 wherever set membership differs. The pairing labels in the first column are abbreviated in the CSV and read as: primary (v1 cohorts) = TCGA-LIHC x GSE57338; HCC array vs HF RNA-seq = GSE76427 x GSE116250; HCC array vs HF RNA-seq 2 = GSE14520_GPL3921 x GSE141910; HCC array vs HF array-adjacent = GSE14520_GPL571 x GSE57338.

Complete table: Table_S18_mirror_summary.csv (8 rows x 13 columns).

pair	tier	n.sets	n.clusters	n.mirror.sets	null.sets mean	p.perm.sets	n.mirror clusters	null.clusters mean	p.perm clusters	p.binom.v1 style	n.eff.nyholt	n.eff.li.ji
primary (v1 cohorts)	A	15	5	10	0.11	9.999e-05	3	0.06	0.0031	1	9.44	5
primary (v1 cohorts)	C	33	19	21	0.23	9.999e-05	14	0.18	9.999e-05	0.06357	32.37	15
HCC array vs HF RNA-seq	A	14	5	2	0.79	0.1744	1	0.39	0.2705	0.375	9.18	5.5
HCC array vs HF RNA-seq	C	28	19	5	1.61	0.09339	4	1.26	0.08679	0.01921	30.98	16.5
HCC array vs HF RNA-seq 2	A	14	5	0	0.01	1	0	0.01	1	0.0625	8.68	5
HCC array vs HF RNA-seq 2	C	27	18	11	0.03	0.0002	10	0.02	9.999e-05	0.8145	27.03	14
HCC array vs HF array-adjacent	A	15	5	9	0.31	0.0014	3	0.18	0.009099	1	9.39	5.5
HCC array vs HF array-adjacent	C	29	19	19	0.61	0.0002	14	0.5	9.999e-05	0.06357	28.8	13

S19 Table (page 1 of 2)

Module preservation, all modules. Zsummary and median rank for every named module in both directions, 200 permutations. The legend states that modulePreservation subsamples modules larger than 1,000 genes, so the three largest modules per direction are computed on a random 1,000-gene subsample. Unassigned (grey) modules appear as rows in the table for completeness and are reported where relevant (the HF → HCC grey module reaches Zsummary = 10.76), but they are excluded from the named-module counts quoted in the main text.

reference	test	module	moduleSize	Zsummary.pres	medianRank pres	preservation class
HCC	HF	magenta	146	7.094	5	weak-moderate (2-10)
HCC	HF	purple	145	2.913	16	weak-moderate (2-10)
HF	HCC	purple	153	11.03	3	strong (>10)
HF	HCC	magenta	167	-1.068	23	none (<2)
HCC	HF	red	302	15.01	3	strong (>10)
HCC	HF	greenyellow	130	12.31	1	strong (>10)
HCC	HF	yellow	347	8.89	6	weak-moderate (2-10)
HCC	HF	lightcyan	90	8.752	2	weak-moderate (2-10)
HCC	HF	turquoise	1000	7.961	9	weak-moderate (2-10)
HCC	HF	brown	389	5.221	10	weak-moderate (2-10)
HCC	HF	green	323	4.812	11	weak-moderate (2-10)
HCC	HF	midnightblue	96	4.722	7	weak-moderate (2-10)
HCC	HF	black	255	3.784	13	weak-moderate (2-10)
HCC	HF	royalblue	43	2.807	6	weak-moderate (2-10)
HCC	HF	cyan	99	2.457	10	weak-moderate (2-10)
HCC	HF	pink	254	2.347	15	weak-moderate (2-10)
HCC	HF	blue	1000	1.048	18	none (<2)
HCC	HF	grey	1000	0.839	19	none (<2)
HCC	HF	lightyellow	57	0.7656	14	none (<2)
HCC	HF	lightgreen	68	0.7004	17	none (<2)
HCC	HF	salmon	110	0.6828	16	none (<2)
HCC	HF	grey60	78	-0.08104	19	none (<2)

S19 Table (page 2 of 2)

Module preservation, all modules. Zsummary and median rank for every named module in both directions, 200 permutations. The legend states that modulePreservation subsamples modules larger than 1,000 genes, so the three largest modules per direction are computed on a random 1,000-gene subsample. Unassigned (grey) modules appear as rows in the table for completeness and are reported where relevant (the HF → HCC grey module reaches Zsummary = 10.76), but they are excluded from the named-module counts quoted in the main text.

Printed from Table_S19_module_preservation.csv (45 rows x 37 columns): 7 of 37 columns and 44 of 45 rows. Rows cited in the manuscript are printed first. The CSV is the complete and authoritative version.

reference	test	module	moduleSize	Zsummary.pres	medianRank pres	preservation class
HCC	HF	tan	117	-0.8069	20	none (<2)
HF	HCC	green	425	11.51	6	strong (>10)
HF	HCC	grey	1000	10.76	14	strong (>10)
HF	HCC	pink	173	10.59	4	strong (>10)
HF	HCC	turquoise	1000	9.294	10	weak-moderate (2-10)
HF	HCC	black	184	8.377	5	weak-moderate (2-10)
HF	HCC	midnightblue	93	5.585	3	weak-moderate (2-10)
HF	HCC	royalblue	52	4.769	2	weak-moderate (2-10)
HF	HCC	blue	1000	2.863	15	weak-moderate (2-10)
HF	HCC	salmon	105	2.478	10	weak-moderate (2-10)
HF	HCC	lightyellow	58	2.386	7	weak-moderate (2-10)
HF	HCC	darkturquoise	39	2.222	9	weak-moderate (2-10)
HF	HCC	lightgreen	70	2.022	8	weak-moderate (2-10)
HF	HCC	yellow	496	1.72	16	none (<2)
HF	HCC	brown	1000	1.103	19	none (<2)
HF	HCC	grey60	70	0.9942	16	none (<2)
HF	HCC	cyan	103	0.9133	16	none (<2)
HF	HCC	darkred	43	0.8175	11	none (<2)
HF	HCC	darkgreen	41	0.6741	14	none (<2)
HF	HCC	tan	146	0.2566	17	none (<2)
HF	HCC	greenyellow	150	0.192	18	none (<2)
HF	HCC	red	189	0.1829	19	none (<2)

S20 Table (panel a)

Module preservation, designated modules. (a) Zsummary, median rank, BH FDR, rank among named modules, and preservation class for HF purple tested in HCC and HCC magenta tested in HF. (b) The two-row verdict summary in the form used in the Results text.

Complete table: Table_S20_designated_module_preservation.csv (2 rows x 37 columns).

reference	module	tested	moduleSize	Zsummary	Zdensity.pres	Zconnectivity.pres	Z.propVarExpla.pres	Z.meanExpla.pres	Z.meanSignAware.pres	Z.meanSignAwareCor.pres	Z.mean.pres	Z.meanAdj.pres	Z.cor.kME	Z.cor.cor	Z.cor.kIM	Z.cor.MS	Z.cor.MEDiss	moduleSize.obs	medianRank.pres	medianRankDensity.pres	medianRankConnectivity.pres	medianRankCombined.pres	propVarExpla.pres	meanExpla.pres	meanSignAware.pres	meanSignAwareCor.pres	meanAdj.pres	medianMAR.pres	cor.kME.pres	cor.kIM.pres	cor.kME.obs	cor.cor.pres	BM.pres	log.p.BHfdr.pres	log.p.Bonf.pres	log.p.fdr.connectivity.pres	p.sup.pres	p.sup.obs	Zsummary.pres	preservation class
HCC	mag	HF	146	7.094	10.35	3.842	5.852	6.455	14.74	14.23	3.643	3.135	3.842	12.39	9.956	4.241	146	5	4	7	0.2505	0.3558	0.1396	0.008446	0.03157	0.7268	0.1421	0.09977	0.1988	0.8534	-14.72	-26.57	-2.87e-14	1.704e-15	1.995e-15	7.094	weak-moderate
HF	pur	HC	153	11.03	17.78	4.27	10.85	10.22	24.71	44	20.65	4.273	3.692	28.21	9.843	3.757	153	3	2	4	0.5479	0.557	0.3213	0.1026	0.2107	0.7358	0.2433	0.2037	0.1907	0.7228	-41.46	-79.3	-3.51e-41	5.194e-42	4.293e-41	11.03	strong (>10)

S20 Table (panel b)

Module preservation, designated modules. (a) Zsummary, median rank, BH FDR, rank among named modules, and preservation class for HF purple tested in HCC and HCC magenta tested in HF. (b) The two-row verdict summary in the form used in the Results text.

Complete table: Table_S20b_designated_module_verdict.csv (2 rows x 5 columns).

reference	module	test	Zsummary	verdict
HCC	magenta	HF	7.094	weakly preserved
HF	purple	HCC	11.03	strongly preserved

S21 Table (page 1 of 2)

Between-cohort heterogeneity. Cochran's Q, PQ, I^2, and pooled effect size with standard error for every set within each disease.

disease	set	tier	n.cohorts	Q	p.Q	I2.pct	g.pooled	se.pooled	cohorts
HCC	...	A	4	51.83	3.255e-11	94.2	1.618	0.3937	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	A	4	49.8	8.794e-11	94	1.485	0.3862	TCGA_LIHC; GSE14520_GPL3921;...
HCC	REACTOME_TRANSLATION	A	4	103.2	3.251e-22	97.1	1.472	0.5056	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	A	4	29.02	2.213e-06	89.7	1.298	0.2946	TCGA_LIHC; GSE14520_GPL3921;...
HCC	REACTOME_RRNA_PROCESSING	A	4	36.61	5.566e-08	91.8	1.264	0.2481	TCGA_LIHC; GSE14520_GPL3921;...
HCC	REACTOME_TRNA_AMINOACYLATION	A	4	5.957	0.1137	49.6	1.247	0.1064	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	A	2	0.05818	0.8094	0	1.18	0.1155	TCGA_LIHC; GSE14520_GPL571
HCC	...	A	4	30.94	8.767e-07	90.3	1.15	0.2631	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	A	4	4.487	0.2135	33.1	0.9885	0.09697	TCGA_LIHC; GSE14520_GPL3921;...
HCC	KEGG_RIBOSOME	A	4	20.42	0.000139	85.3	0.9477	0.215	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	A	4	60.86	3.847e-13	95.1	0.8647	0.3536	TCGA_LIHC; GSE14520_GPL3921;...
HCC	REACTOME_MITOCHONDRIAL_TRANSLATION	A	4	68.77	7.823e-15	95.6	0.8507	0.373	TCGA_LIHC; GSE14520_GPL3921;...
HCC	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	4	11.17	0.01084	73.1	0.8458	0.1395	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	A	4	13.76	0.003257	78.2	0.6806	0.155	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	A	4	16.78	0.0007854	82.1	-0.2118	0.1556	TCGA_LIHC; GSE14520_GPL3921;...
HF	...	A	3	11.3	0.003524	82.3	-0.7436	0.2037	GSE57338; GSE116250; GSE141910
HF	...	A	3	0.2268	0.8928	0	0.5241	0.07572	GSE57338; GSE116250; GSE141910
HF	REACTOME_RRNA_PROCESSING	A	3	31.84	1.22e-07	93.7	-0.3074	0.3201	GSE57338; GSE116250; GSE141910
HF	...	A	3	6.473	0.03931	69.1	-0.2856	0.1432	GSE57338; GSE116250; GSE141910
HF	REACTOME_TRANSLATION	A	3	16.98	0.0002056	88.2	-0.2826	0.2445	GSE57338; GSE116250; GSE141910
HF	REACTOME_MITOCHONDRIAL_TRANSLATION	A	3	9.682	0.0079	79.3	-0.1982	0.1838	GSE57338; GSE116250; GSE141910
HF	...	A	3	28.95	5.162e-07	93.1	-0.1739	0.3325	GSE57338; GSE116250; GSE141910

S21 Table (page 2 of 2)

Between-cohort heterogeneity. Cochran's Q, PQ, I^2, and pooled effect size with standard error for every set within each disease.

Printed from Table_S21_cohort_heterogeneity.csv (82 rows x 10 columns): 44 of 82 rows. Rows cited in the manuscript are printed first. The CSV is the complete and authoritative version.

disease	set	tier	n.cohorts	Q	p.Q	I2.pct	g.pooled	se.pooled	cohorts
HF	...	A	3	8.333	0.0155	76	-0.1355	0.1697	GSE57338; GSE116250; GSE141910
HF	...	A	3	18.18	0.0001129	89	-0.1277	0.2668	GSE57338; GSE116250; GSE141910
HF	REACTOME_TRNA_AMINOACYLATION	A	3	7.223	0.02701	72.3	-0.1125	0.1565	GSE57338; GSE116250; GSE141910
HF	...	A	3	26.95	1.403e-06	92.6	-0.07609	0.3231	GSE57338; GSE116250; GSE141910
HF	...	A	3	26.62	1.658e-06	92.5	0.0533	0.3303	GSE57338; GSE116250; GSE141910
HF	...	A	3	28.83	5.494e-07	93.1	-0.04594	0.3434	GSE57338; GSE116250; GSE141910
HF	KEGG_RIBOSOME	A	3	24.03	6.065e-06	91.7	0.01552	0.309	GSE57338; GSE116250; GSE141910
HF	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	3	4.976	0.08306	59.8	-0.002771	0.1301	GSE57338; GSE116250; GSE141910
HCC	...	B_extra	4	67.68	1.338e-14	95.6	2.255	0.4344	TCGA_LIHC; GSE14520_GPL3921;...
HCC	REACTOME_TRNA_PROCESSING	B_extra	4	59.91	6.132e-13	95	2.099	0.4104	TCGA_LIHC; GSE14520_GPL3921;...
HCC	HALLMARK_MYC_TARGETS_V1	C_v1	4	71.97	1.618e-15	95.8	2.065	0.4062	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	C_v1	4	15.87	0.001207	81.1	1.662	0.2013	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	C_v1	4	25	1.543e-05	88	1.627	0.2879	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	C_v1	4	31.94	5.398e-07	90.6	1.463	0.2888	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	C_v1	4	78.04	8.072e-17	96.2	1.451	0.4348	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	C_v1	4	21.28	9.224e-05	85.9	1.377	0.2171	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	B_extra	4	20.71	0.0001207	85.5	1.213	0.2154	TCGA_LIHC; GSE14520_GPL3921;...
HCC	HALLMARK_MYC_TARGETS_V2	C_v1	4	28.78	2.488e-06	89.6	1.136	0.227	TCGA_LIHC; GSE14520_GPL3921;...
HCC	HALLMARK_MTORC1_SIGNALING	C_v1	4	72.66	1.15e-15	95.9	1.119	0.3813	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	C_v1	4	11.81	0.008056	74.6	1.098	0.1521	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	B_extra	4	27.56	4.502e-06	89.1	1.092	0.2161	TCGA_LIHC; GSE14520_GPL3921;...
HCC	...	C_v1	4	9.17	0.02711	67.3	0.6775	0.1263	TCGA_LIHC; GSE14520_GPL3921;...

S22 Table (panel a) (page 1 of 2)

HF aetiology. (a) Stratification summary: each HF subgroup against TCGA-LIHC, with subgroup sizes, cross-disease Spearman rho, mirror set counts before and after the effect-size gate, and mean g. (b) Per-set Hedges g for each aetiology subgroup, with a flag for whether the DCM and ICM arms of GSE57338 agree in direction. (c) Cross-disease correlations computed separately within each aetiology arm, so that aetiology composition can be assessed as an explanation for cohort discordance. (d) The same aetiology (dilated cardiomyopathy) sampled in GSE57338 and GSE141910 and compared per set, showing the same-aetiology, cross-cohort discordance.

set	g.DCM.vs.NF	g.ICM.vs.NF	tier	same direction
KEGG_RIBOSOME	-0.2746	-0.5563	A	True
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	-0.207	-0.2082	A	True
REACTOME_TRANSLATION	-0.4619	-0.6029	A	True
...	-0.4714	-0.7269	A	True
...	-0.2922	-0.5914	A	True
REACTOME_TRNA_AMINOACYLATION	-0.3564	-0.3557	A	True
...	-0.8103	-0.8461	A	True
REACTOME_RRNA_PROCESSING	-0.6626	-0.7588	A	True
...	-0.4564	-0.6578	A	True
...	-0.3782	-0.7275	A	True
...	-0.3541	-0.6801	A	True
REACTOME_MITOCHONDRIAL_TRANSLATION	-0.05858	-0.1123	A	True
...	-0.05923	-0.0992	A	True
...	0.5985	0.5506	A	True
...	-0.1222	-0.09263	A	True
REACTOME_TRNA_PROCESSING	-0.4041	-0.4249	B_extra	True
...	-0.4822	-0.5813	B_extra	True
...	-0.1847	-0.08593	B_extra	True
...	-0.7435	-0.76	B_extra	True
...	-0.4767	-0.4773	B_extra	True
...	-0.3308	-0.4195	B_extra	True
...	-0.5631	-0.6587	C_v1	True

S22 Table (panel a) (page 2 of 2)

HF aetiology. (a) Stratification summary: each HF subgroup against TCGA-LIHC, with subgroup sizes, cross-disease Spearman rho, mirror set counts before and after the effect-size gate, and mean g. (b) Per-set Hedges g for each aetiology subgroup, with a flag for whether the DCM and ICM arms of GSE57338 agree in direction. (c) Cross-disease correlations computed separately within each aetiology arm, so that aetiology composition can be assessed as an explanation for cohort discordance. (d) The same aetiology (dilated cardiomyopathy) sampled in GSE57338 and GSE141910 and compared per set, showing the same-aetiology, cross-cohort discordance.

Complete table: Table_S22_HF_aetiology_stratified.csv (41 rows x 5 columns).

set	g.DCM.vs.NF	g.ICM.vs.NF	tier	same direction
HALLMARK_MYC_TARGETS_V1	-0.7464	-0.7666	C_v1	True
...	-0.5091	-0.6278	C_v1	True
...	-0.6841	-0.7331	C_v1	True
...	-0.9324	-1.031	C_v1	True
HALLMARK_MYC_TARGETS_V2	-1.098	-1.01	C_v1	True
...	-0.4667	-0.7082	C_v1	True
...	-1.013	-1.179	C_v1	True
...	-1.126	-1.142	C_v1	True
...	-0.3616	-0.6411	C_v1	True
HALLMARK_MTORC1_SIGNALING	-1.066	-0.8098	C_v1	True
...	-0.06269	0.1354	C_v1	False
...	-0.2431	-0.3304	C_v1	True
...	-0.8361	-0.3439	C_v1	True
...	-1.548	-1.683	C_v1	True
...	0.4368	0.1951	C_v1	True
...	-1.133	-1.041	C_v1	True
...	0.2685	0.03703	C_v1	True
...	0.3304	0.07489	C_v1	True
...	0.06916	-0.217	C_v1	False

S22 Table (panel b) (page 1 of 2)

HF aetiology. (a) Stratification summary: each HF subgroup against TCGA-LIHC, with subgroup sizes, cross-disease Spearman rho, mirror set counts before and after the effect-size gate, and mean g. (b) Per-set Hedges g for each aetiology subgroup, with a flag for whether the DCM and ICM arms of GSE57338 agree in direction. (c) Cross-disease correlations computed separately within each aetiology arm, so that aetiology composition can be assessed as an explanation for cohort discordance. (d) The same aetiology (dilated cardiomyopathy) sampled in GSE57338 and GSE141910 and compared per set, showing the same-aetiology, cross-cohort discordance.

cohort	aetiology	n.aetiology	n.control	set	tier	hedges.g	direction
GSE57338	DCM	82	136	KEGG_RIBOSOME	A	-0.2746	down in disease
GSE57338	DCM	82	136	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	-0.207	down in disease
GSE57338	DCM	82	136	REACTOME_TRANSLATION	A	-0.4619	down in disease
GSE57338	DCM	82	136	...	A	-0.4714	down in disease
GSE57338	DCM	82	136	...	A	-0.2922	down in disease
GSE57338	DCM	82	136	REACTOME_TRNA_AMINOACYLATION	A	-0.3564	down in disease
GSE57338	DCM	82	136	...	A	-0.8103	down in disease
GSE57338	DCM	82	136	REACTOME_RRNA_PROCESSING	A	-0.6626	down in disease
GSE57338	DCM	82	136	...	A	-0.4564	down in disease
GSE57338	DCM	82	136	...	A	-0.3782	down in disease
GSE57338	DCM	82	136	...	A	-0.3541	down in disease
GSE57338	DCM	82	136	REACTOME_MITOCHONDRIAL_TRANSLATION	A	-0.05858	down in disease
GSE57338	DCM	82	136	...	A	-0.05923	down in disease
GSE57338	DCM	82	136	...	A	0.5985	up in disease
GSE57338	DCM	82	136	...	A	-0.1222	down in disease
GSE57338	ICM	95	136	KEGG_RIBOSOME	A	-0.5563	down in disease
GSE57338	ICM	95	136	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	-0.2082	down in disease
GSE57338	ICM	95	136	REACTOME_TRANSLATION	A	-0.6029	down in disease
GSE57338	ICM	95	136	...	A	-0.7269	down in disease
GSE57338	ICM	95	136	...	A	-0.5914	down in disease
GSE57338	ICM	95	136	REACTOME_TRNA_AMINOACYLATION	A	-0.3557	down in disease
GSE57338	ICM	95	136	...	A	-0.8461	down in disease

S22 Table (panel b) (page 2 of 2)

HF aetiology. (a) Stratification summary: each HF subgroup against TCGA-LIHC, with subgroup sizes, cross-disease Spearman rho, mirror set counts before and after the effect-size gate, and mean g. (b) Per-set Hedges g for each aetiology subgroup, with a flag for whether the DCM and ICM arms of GSE57338 agree in direction. (c) Cross-disease correlations computed separately within each aetiology arm, so that aetiology composition can be assessed as an explanation for cohort discordance. (d) The same aetiology (dilated cardiomyopathy) sampled in GSE57338 and GSE141910 and compared per set, showing the same-aetiology, cross-cohort discordance.

Printed from Table_S22b_HF_aetiology_effect_sizes.csv (75 rows x 8 columns): 44 of 75 rows. The CSV is the complete and authoritative version.

cohort	aetiology	n.aetiology	n.control	set	tier	hedges.g	direction
GSE57338	ICM	95	136	REACTOME_RRNA_PROCESSING	A	-0.7588	down in disease
GSE57338	ICM	95	136	...	A	-0.6578	down in disease
GSE57338	ICM	95	136	...	A	-0.7275	down in disease
GSE57338	ICM	95	136	...	A	-0.6801	down in disease
GSE57338	ICM	95	136	REACTOME_MITOCHONDRIAL_TRANSLATION	A	-0.1123	down in disease
GSE57338	ICM	95	136	...	A	-0.0992	down in disease
GSE57338	ICM	95	136	...	A	0.5506	up in disease
GSE57338	ICM	95	136	...	A	-0.09263	down in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	KEGG_RIBOSOME	A	0.3306	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	0.1561	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	REACTOME_TRANSLATION	A	0.09562	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	...	A	0.2051	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	...	A	0.3325	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	REACTOME_TRNA_AMINOACYLATION	A	0.07557	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	...	A	-0.4163	down in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	REACTOME_RRNA_PROCESSING	A	-0.005967	down in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	...	A	0.2235	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	...	A	0.191	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	...	A	0.09461	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	REACTOME_MITOCHONDRIAL_TRANSLATION	A	0.1033	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	...	A	0.1514	up in disease
GSE141910	Dilated cardiomyopathy (DCM)	166	166	...	A	0.5812	up in disease

S22 Table (panel c)

HF aetiology. (a) Stratification summary: each HF subgroup against TCGA-LIHC, with subgroup sizes, cross-disease Spearman rho, mirror set counts before and after the effect-size gate, and mean g. (b) Per-set Hedges g for each aetiology subgroup, with a flag for whether the DCM and ICM arms of GSE57338 agree in direction. (c) Cross-disease correlations computed separately within each aetiology arm, so that aetiology composition can be assessed as an explanation for cohort discordance. (d) The same aetiology (dilated cardiomyopathy) sampled in GSE57338 and GSE141910 and compared per set, showing the same-aetiology, cross-cohort discordance.

Complete table: Table_S22c_HF_aetiology_vs_HCC.csv (5 rows x 10 columns).

cohort	aetiology	n.aetiology	n.control	n.sets	spearman.rho vs.HCC	mirror.sets abs.g.ge.020	mirror.sets gated	n.gated	mean.g
GSE141910	Dilated cardiomyopathy (DCM)	166	166	15	-0.1929	1	1	13	0.1319
GSE141910	Peripartum cardiomyopathy (PPCM)	6	166	15	0.1143	3	3	13	-0.1263
GSE141910	Hypertrophic cardiomyopathy (HCM)	28	166	15	-0.1	2	2	13	-0.01946
GSE57338	ICM	95	136	15	-0.5964	11	11	13	-0.431
GSE57338	DCM	82	136	15	-0.5536	11	11	13	-0.2911

S22 Table (panel d)

HF aetiology. (a) Stratification summary: each HF subgroup against TCGA-LIHC, with subgroup sizes, cross-disease Spearman rho, mirror set counts before and after the effect-size gate, and mean g. (b) Per-set Hedges g for each aetiology subgroup, with a flag for whether the DCM and ICM arms of GSE57338 agree in direction. (c) Cross-disease correlations computed separately within each aetiology arm, so that aetiology composition can be assessed as an explanation for cohort discordance. (d) The same aetiology (dilated cardiomyopathy) sampled in GSE57338 and GSE141910 and compared per set, showing the same-aetiology, cross-cohort discordance.

Complete table: Table_S22d_DCM_across_cohorts.csv (15 rows x 3 columns).

set	g.GSE57338 DCM	g.GSE141910 DCM
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	-0.207	0.1561
KEGG_RIBOSOME	-0.2746	0.3306
...	-0.8103	-0.4163
...	-0.2922	0.3325
...	-0.4714	0.2051
REACTOME_MITOCHONDRIAL_TRANSLATION	-0.05858	0.1033
...	-0.05923	0.1514
...	0.5985	0.5812
...	-0.3782	0.191
...	-0.3541	0.09461
REACTOME_RRNA_PROCESSING	-0.6626	-0.005967
...	-0.1222	-0.1394
...	-0.4564	0.2235
REACTOME_TRANSLATION	-0.4619	0.09562
REACTOME_TRNA_AMINOACYLATION	-0.3564	0.07557

S23 Table (page 1 of 2)

Composition marker coverage. The marker genes available in each cohort for each composition panel, with the panels lost entirely on the GSE57338 array made explicit — 10 of 24 pan-immune markers are retained, and the lymphoid and proliferation panels are lost outright, so that cohort's "immune-adjusted" model is myeloid-only.

cohort	panel	n.used	n.total	frac.used	mean.rank	mean.z	missing
GSE57338	immune_pan	10	24	0.417	4699	1.836e-17	...
GSE57338	myeloid	7	9	0.778	6445	-2.781e-17	FCGR3A,ITGAX
GSE57338	fibroblast	12	14	0.857	8480	6.298e-16	FAP,THY1
GSE57338	endothelial	8	8	1	8504	-1.035e-16	
GSE57338	hepatocyte	3	14	0.214	3871	-1.382e-16	...
GSE57338	cardiomyocyte	14	14	1	1.121e+04	2.348e-16	
GSE141198	immune_pan	19	24	0.792	5036	1.223e-17	CD3G,CD19,MS4A1,CD68,IL2RA
GSE141198	myeloid	8	9	0.889	7613	-1.144e-16	CD68
GSE141198	lymphocyte	8	10	0.8	3771	1.07e-16	CD19,MS4A1
GSE141198	fibroblast	11	14	0.786	7927	5.157e-17	LUM,FAP,POSTN
GSE141198	endothelial	8	8	1	7502	3.277e-17	
GSE141198	hepatocyte	14	14	1	1.271e+04	-3.124e-17	
GSE141198	cardiomyocyte	3	14	0.214	5105	-1.594e-16	...
GSE141198	proliferation	11	11	1	5475	-1.427e-16	
TCGA_LIHC	immune_pan	23	24	0.958	7512	-7.409e-18	CD68
TCGA_LIHC	myeloid	8	9	0.889	1.124e+04	-8.18e-17	CD68
TCGA_LIHC	lymphocyte	10	10	1	5991	7.239e-17	
TCGA_LIHC	fibroblast	14	14	1	1.2e+04	1.316e-16	
TCGA_LIHC	endothelial	8	8	1	1.202e+04	9.696e-18	
TCGA_LIHC	hepatocyte	14	14	1	1.637e+04	-8.279e-17	
TCGA_LIHC	cardiomyocyte	4	14	0.286	6527	-2.027e-16	...
TCGA_LIHC	proliferation	11	11	1	9213	1.261e-16	

S23 Table (page 2 of 2)

Composition marker coverage. The marker genes available in each cohort for each composition panel, with the panels lost entirely on the GSE57338 array made explicit — 10 of 24 pan-immune markers are retained, and the lymphoid and proliferation panels are lost outright, so that cohort's "immune-adjusted" model is myeloid-only.

Printed from Table_S23_composition_marker_coverage.csv (60 rows x 8 columns): 44 of 60 rows. The CSV is the complete and authoritative version.

cohort	panel	n.used	n.total	frac.used	mean.rank	mean.z	missing
GSE116250	immune_pan	24	24	1	1.215e+04	7.571e-18	
GSE116250	myeloid	9	9	1	1.819e+04	1.301e-17	
GSE116250	lymphocyte	10	10	1	9689	-3.753e-18	
GSE116250	fibroblast	14	14	1	2.316e+04	-2.038e-17	
GSE116250	endothelial	7	8	0.875	2.167e+04	-1.493e-17	PECAM1
GSE116250	hepatocyte	9	14	0.643	1.123e+04	-4.857e-17	CYP3A4,FGA,FGB,FGG,APOC3
GSE116250	cardiomyocyte	14	14	1	2.594e+04	2.835e-17	
GSE116250	proliferation	11	11	1	1.058e+04	3.523e-18	
GSE14520_GPL3921	immune_pan	11	24	0.458	3502	1.67e-16	...
GSE14520_GPL3921	myeloid	5	9	0.556	3817	2.984e-16	CD68,FCGR3A,LYZ,ITGAX
GSE14520_GPL3921	lymphocyte	4	10	0.4	2172	6.282e-17	CD3E,CD19,MS4A1,NKG7,GNLY,KLRD1
GSE14520_GPL3921	fibroblast	13	14	0.929	4271	-5.479e-17	FAP
GSE14520_GPL3921	endothelial	7	8	0.875	2139	-1.149e-17	TEK
GSE14520_GPL3921	hepatocyte	14	14	1	7379	-1.547e-16	
GSE14520_GPL3921	proliferation	4	11	0.364	3759	-2.159e-17	...
GSE14520_GPL571	immune_pan	12	24	0.5	3480	-4.649e-17	...
GSE14520_GPL571	myeloid	5	9	0.556	4757	1.232e-18	CD68,FCGR3A,ITGAM,ITGAX
GSE14520_GPL571	lymphocyte	5	10	0.5	1871	-2.372e-17	CD19,MS4A1,NKG7,GNLY,KLRD1
GSE14520_GPL571	fibroblast	13	14	0.929	4435	-5.566e-17	FAP
GSE14520_GPL571	endothelial	7	8	0.875	2590	6.362e-17	TEK
GSE14520_GPL571	hepatocyte	14	14	1	7403	-9.841e-17	
GSE14520_GPL571	proliferation	5	11	0.455	3510	-9.904e-17	MKI67,CCNB1,CDK1,BUB1,PLK1,CCNA2

S24 Table (page 1 of 2)

Composition shift. Marker-based composition scores by group in each cohort, showing the direction and magnitude of the composition difference between disease and control arms.

cohort	disease	panel	n.disease	n.control	mean.disease	mean.control	hedges.g	p.ttest	p.fdr
GSE57338	HF	immune_pan	177	136	4527	4923	-0.3597	0.001153	0.001383
GSE57338	HF	myeloid	177	136	6199	6765	-0.4904	1.184e-05	7.105e-05
GSE57338	HF	fibroblast	177	136	8645	8266	0.445	0.0001025	0.0002511
GSE57338	HF	endothelial	177	136	8448	8575	-0.3728	0.0009568	0.001383
GSE57338	HF	hepatocyte	177	136	3706	4086	-0.4323	0.0001256	0.0002511
GSE57338	HF	cardiomyocyte	177	136	1.129e+04	1.11e+04	0.3223	0.003833	0.003833
TCGA_LIHC	HCC	immune_pan	371	50	7419	8204	-0.4506	0.0002319	0.0003092
TCGA_LIHC	HCC	myeloid	371	50	1.113e+04	1.208e+04	-0.5322	2.787e-05	4.459e-05
TCGA_LIHC	HCC	lymphocyte	371	50	5855	7005	-0.5972	4.992e-08	9.984e-08
TCGA_LIHC	HCC	fibroblast	371	50	1.198e+04	1.217e+04	-0.08356	0.2665	0.2665
TCGA_LIHC	HCC	endothelial	371	50	1.195e+04	1.248e+04	-0.324	0.001166	0.001333
TCGA_LIHC	HCC	hepatocyte	371	50	1.627e+04	1.712e+04	-0.5822	5.993e-23	2.397e-22
TCGA_LIHC	HCC	cardiomyocyte	371	50	6603	5966	0.6018	8.247e-14	2.199e-13
TCGA_LIHC	HCC	proliferation	371	50	9791	4922	2.227	3.36e-47	2.688e-46
GSE116250	HF	immune_pan	50	14	1.234e+04	1.149e+04	0.5469	0.03318	0.04424
GSE116250	HF	myeloid	50	14	1.811e+04	1.844e+04	-0.2004	0.4697	0.4697
GSE116250	HF	lymphocyte	50	14	1.002e+04	8493	0.9507	0.0001904	0.0007617
GSE116250	HF	fibroblast	50	14	2.352e+04	2.189e+04	1.717	0.0004642	0.001238
GSE116250	HF	endothelial	50	14	2.183e+04	2.108e+04	1.341	5.045e-07	4.036e-06
GSE116250	HF	hepatocyte	50	14	1.117e+04	1.141e+04	-0.2595	0.4187	0.4697
GSE116250	HF	cardiomyocyte	50	14	2.604e+04	2.561e+04	1.674	0.002228	0.004456
GSE116250	HF	proliferation	50	14	1.027e+04	1.166e+04	-1.266	0.01355	0.02168

S24 Table (page 2 of 2)

Composition shift. Marker-based composition scores by group in each cohort, showing the direction and magnitude of the composition difference between disease and control arms.

Printed from Table_S24_composition_shift.csv (52 rows x 10 columns): 44 of 52 rows. The CSV is the complete and authoritative version.

cohort	disease	panel	n.disease	n.control	mean.disease	mean.control	hedges.g	p.ttest	p.fdr
GSE14520_GPL3921	HCC	immune_pan	225	220	3135	3877	-0.8265	5.764e-17	8.069e-17
GSE14520_GPL3921	HCC	myeloid	225	220	3478	4164	-0.9374	5.975e-21	1.046e-20
GSE14520_GPL3921	HCC	lymphocyte	225	220	1711	2644	-0.9854	9.099e-23	3.185e-22
GSE14520_GPL3921	HCC	fibroblast	225	220	4059	4487	-0.4215	1.014e-05	1.183e-05
GSE14520_GPL3921	HCC	endothelial	225	220	2080	2200	-0.181	0.05597	0.05597
GSE14520_GPL3921	HCC	hepatocyte	225	220	7198	7564	-0.9779	1.149e-21	2.68e-21
GSE14520_GPL3921	HCC	proliferation	225	220	5130	2357	3.017	7.528e-111	5.269e-110
GSE14520_GPL571	HCC	immune_pan	22	19	3171	3837	-0.8078	0.009723	0.01702
GSE14520_GPL571	HCC	myeloid	22	19	4390	5183	-1.026	0.001421	0.003316
GSE14520_GPL571	HCC	lymphocyte	22	19	1615	2166	-0.7622	0.01396	0.01955
GSE14520_GPL571	HCC	fibroblast	22	19	4332	4555	-0.1795	0.5398	0.6005
GSE14520_GPL571	HCC	endothelial	22	19	2642	2528	0.1546	0.6005	0.6005
GSE14520_GPL571	HCC	hepatocyte	22	19	7170	7671	-1.147	0.0006099	0.002135
GSE14520_GPL571	HCC	proliferation	22	19	4978	1811	3.116	1.15e-12	8.05e-12
GSE76427	HCC	immune_pan	115	52	7718	8788	-0.6595	1.921e-05	3.074e-05
GSE76427	HCC	myeloid	115	52	1.082e+04	1.173e+04	-0.6432	1.531e-05	3.062e-05
GSE76427	HCC	lymphocyte	115	52	6086	8046	-0.8249	1.111e-07	2.964e-07
GSE76427	HCC	fibroblast	115	52	8705	9675	-0.4353	0.002453	0.002804
GSE76427	HCC	endothelial	115	52	9027	9877	-0.5002	0.001072	0.001429
GSE76427	HCC	hepatocyte	115	52	1.628e+04	1.681e+04	-0.8143	3.714e-10	1.486e-09
GSE76427	HCC	cardiomyocyte	115	52	4497	4719	-0.2888	0.04875	0.04875
GSE76427	HCC	proliferation	115	52	9717	6465	1.693	1.525e-22	1.22e-21

S25 Table (columns 1 of 2)

Composition adjustment. The unadjusted and adjusted correlation, the attenuation, whether the direction survives adjustment, and the maximum variance inflation factor, for every set–cohort pair across all three tiers (268 rows: 103 Tier A, 38 Tier B-extra, 127 Tier C-v1, over seven cohorts). The summary quoted in the main text — 55 of 103 pairs survive, median attenuation 34.1%, maximum VIF 1.94 — is computed on the Tier A subset alone; the Tier B and C rows are provided for completeness and are not part of that summary.

Printed from Table_S25_composition_adjustment.csv (268 rows x 20 columns): 22 of 268 rows. The CSV is the complete and authoritative version.

cohort	disease	set	tier	n	g.unadjusted	p.unadjusted	g.adjusted	p.adjusted	n.covariates	max.vif	g.one.at.a time.worst
GSE57338	HF	KEGG_RIBOSOME	A	313	-0.4473	0.000108	-0.4774	9.783e-05	3	1.388	-0.2937
GSE57338	HF	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	313	-0.2085	0.06849	-0.2253	0.06345	3	1.388	-0.4726
GSE57338	HF	REACTOME_TRANSLATION	A	313	-0.539	3.462e-06	-0.4611	0.0001663	3	1.388	-0.4142
GSE57338	HF	...	A	313	-0.6511	2.644e-08	-0.6786	4.49e-08	3	1.388	-0.539
GSE57338	HF	...	A	313	-0.4757	3.924e-05	-0.5152	2.724e-05	3	1.388	-0.3303
GSE57338	HF	REACTOME_TRNA_AMINOACYLATION	A	313	-0.3621	0.001648	-0.3482	0.004265	3	1.388	-0.2201
GSE57338	HF	...	A	313	-0.9199	1.575e-14	-0.8983	1.095e-12	3	1.388	-0.9995
GSE57338	HF	REACTOME_RRNA_PROCESSING	A	313	-0.8114	7.726e-12	-0.7515	1.681e-09	3	1.388	-0.7821
GSE57338	HF	...	A	313	-0.6071	1.944e-07	-0.6397	2.332e-07	3	1.388	-0.481
GSE57338	HF	...	A	313	-0.5993	2.741e-07	-0.561	5.199e-06	3	1.388	-0.5369
GSE57338	HF	...	A	313	-0.5521	2.034e-06	-0.4676	0.0001347	3	1.388	-0.654
GSE57338	HF	REACTOME_MITOCHONDRIAL_TRANSLATION	A	313	-0.08824	0.4396	0.0285	0.8138	3	1.388	0.2223
GSE57338	HF	...	A	313	-0.08121	0.4769	0.03919	0.7462	3	1.388	0.2132
GSE57338	HF	...	A	313	0.5582	1.578e-06	0.7671	8.05e-10	3	1.388	0.3901
GSE57338	HF	...	A	313	-0.1005	0.3786	-0.04998	0.6797	3	1.388	0.2097
GSE57338	HF	REACTOME_TRNA_PROCESSING	B_extra	313	-0.4411	0.0001333	-0.4269	0.0004795	3	1.388	-0.6091
GSE57338	HF	...	B_extra	313	-0.5808	6.096e-07	-0.5856	2.041e-06	3	1.388	-0.6828
GSE57338	HF	...	B_extra	313	-0.1418	0.2147	-0.1195	0.3239	3	1.388	-0.2624
GSE57338	HF	...	B_extra	313	-0.8652	3.821e-13	-0.8064	1.201e-10	3	1.388	-0.9262
GSE57338	HF	...	B_extra	313	-0.4652	5.745e-05	-0.4201	0.0005874	3	1.388	-0.2729
GSE57338	HF	...	B_extra	313	-0.3714	0.001249	-0.3574	0.003366	3	1.388	-0.23
GSE57338	HF	...	C_v1	313	-0.6502	2.754e-08	-0.9312	1.879e-13	3	1.388	-0.9509

S25 Table (columns 2 of 2)

Composition adjustment. The unadjusted and adjusted correlation, the attenuation, whether the direction survives adjustment, and the maximum variance inflation factor, for every set–cohort pair across all three tiers (268 rows: 103 Tier A, 38 Tier B-extra, 127 Tier C-v1, over seven cohorts). The summary quoted in the main text — 55 of 103 pairs survive, median attenuation 34.1%, maximum VIF 1.94 — is computed on the Tier A subset alone; the Tier B and C rows are provided for completeness and are not part of that summary.

Printed from Table_S25_composition_adjustment.csv (268 rows x 20 columns): 22 of 268 rows. The CSV is the complete and authoritative version.

cohort	worst covariate	p.one.at.a time.worst	covariates	attenuation	sign.kept	survives	p.fdr.unadj	p.fdr.adj
GSE57338	myeloid	0.01892	...	-0.06748	True	True	0.000164	0.0001744
GSE57338	myeloid	0.0001641	...	-0.08072	True	False	0.0842	0.07883
GSE57338	fibroblast	0.0007884	...	0.1446	True	True	6.452e-06	0.0002727
GSE57338	myeloid	1.738e-05	...	-0.04225	True	True	8.067e-08	1.227e-07
GSE57338	myeloid	0.008532	...	-0.08291	True	True	6.435e-05	5.077e-05
GSE57338	fibroblast	0.0702	...	0.03821	True	True	0.002252	0.00564
GSE57338	cardiomyocyte	5.515e-16	...	0.02353	True	True	8.069e-14	4.99e-12
GSE57338	myeloid	4.616e-10	...	0.07386	True	True	2.88e-11	4.922e-09
GSE57338	myeloid	0.0001545	...	-0.05363	True	True	4.981e-07	5.976e-07
GSE57338	myeloid	1.908e-05	...	0.06401	True	True	6.611e-07	1.066e-05
GSE57338	cardiomyocyte	1.042e-07	...	0.153	True	True	3.971e-06	0.0002301
GSE57338	fibroblast	0.07569	...	1.323	False	False	0.4743	0.8556
GSE57338	fibroblast	0.08543	...	1.483	False	False	0.4888	0.8051
GSE57338	myeloid	0.003575	...	-0.3741	True	True	3.235e-06	2.539e-09
GSE57338	fibroblast	0.1046	...	0.5029	True	False	0.4196	0.7532
GSE57338	myeloid	6.471e-07	...	0.03232	True	True	0.0001952	0.0007562
GSE57338	myeloid	1.54e-08	...	-0.008124	True	True	1.316e-06	4.404e-06
GSE57338	myeloid	0.02832	...	0.1573	True	False	0.2445	0.3906
GSE57338	cardiomyocyte	5.114e-14	...	0.0679	True	True	1.741e-12	4.922e-10
GSE57338	fibroblast	0.02468	...	0.0969	True	True	9.059e-05	0.000892
GSE57338	fibroblast	0.05578	...	0.03778	True	True	0.001765	0.004601
GSE57338	fibroblast	3.665e-13	...	-0.4321	True	True	8.067e-08	9.63e-13

S26 Table (page 1 of 2)

Composition–set correlation. The correlation between each composition score and each translation set score, so that a set whose apparent disease association tracks composition rather than translation can be identified.

cohort	disease	set	panel	r	p	p.fdr
GSE57338	HF	KEGG_RIBOSOME	immune_pan	0.43	1.607e-15	1.206e-14
GSE57338	HF	KEGG_RIBOSOME	myeloid	0.4216	6.385e-15	4.105e-14
GSE57338	HF	KEGG_RIBOSOME	fibroblast	0.2595	3.278e-06	1.054e-05
GSE57338	HF	KEGG_RIBOSOME	endothelial	-0.01923	0.7348	0.8064
GSE57338	HF	KEGG_RIBOSOME	hepatocyte	0.1182	0.03658	0.06097
GSE57338	HF	KEGG_RIBOSOME	cardiomyocyte	0.1642	0.003569	0.007833
GSE57338	HF	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	immune_pan	-0.3859	1.488e-12	5.579e-12
GSE57338	HF	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	myeloid	-0.3705	1.277e-11	4.598e-11
GSE57338	HF	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	fibroblast	-0.4028	1.229e-13	5.821e-13
GSE57338	HF	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	endothelial	-0.03548	0.5317	0.6296
GSE57338	HF	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	hepatocyte	-0.0607	0.2843	0.3604
GSE57338	HF	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	cardiomyocyte	0.05599	0.3234	0.4043
GSE57338	HF	REACTOME_TRANSLATION	immune_pan	-0.1542	0.006262	0.01299
GSE57338	HF	REACTOME_TRANSLATION	myeloid	-0.1385	0.01421	0.02611
GSE57338	HF	REACTOME_TRANSLATION	fibroblast	-0.4231	5.023e-15	3.478e-14
GSE57338	HF	REACTOME_TRANSLATION	endothelial	-0.06726	0.2354	0.3116
GSE57338	HF	REACTOME_TRANSLATION	hepatocyte	0.09589	0.09036	0.1479
GSE57338	HF	REACTOME_TRANSLATION	cardiomyocyte	0.08024	0.1567	0.2312
GSE57338	HF	...	immune_pan	0.3952	3.822e-13	1.72e-12
GSE57338	HF	...	myeloid	0.3915	6.633e-13	2.616e-12
GSE57338	HF	...	fibroblast	0.1778	0.001588	0.003971
GSE57338	HF	...	endothelial	0.00808	0.8868	0.9163

S26 Table (page 2 of 2)

Composition–set correlation. The correlation between each composition score and each translation set score, so that a set whose apparent disease association tracks composition rather than translation can be identified.

Printed from Table_S26_composition_set_correlation.csv (765 rows x 7 columns): 44 of 765 rows. The CSV is the complete and authoritative version.

cohort	disease	set	panel	r	p	p.fdr
GSE57338	HF	...	hepatocyte	0.1363	0.01579	0.02841
GSE57338	HF	...	cardiomyocyte	0.1456	0.009874	0.01932
GSE57338	HF	...	immune_pan	0.4211	6.936e-15	4.161e-14
GSE57338	HF	...	myeloid	0.4166	1.439e-14	8.092e-14
GSE57338	HF	...	fibroblast	0.2538	5.453e-06	1.583e-05
GSE57338	HF	...	endothelial	-0.000457	0.9936	0.9936
GSE57338	HF	...	hepatocyte	0.127	0.02463	0.04241
GSE57338	HF	...	cardiomyocyte	0.1759	0.001783	0.004338
GSE57338	HF	REACTOME_TRNA_AMINOACYLATION	immune_pan	-0.2848	2.985e-07	1.033e-06
GSE57338	HF	REACTOME_TRNA_AMINOACYLATION	myeloid	-0.2709	1.144e-06	3.813e-06
GSE57338	HF	REACTOME_TRNA_AMINOACYLATION	fibroblast	-0.3927	5.567e-13	2.386e-12
GSE57338	HF	REACTOME_TRNA_AMINOACYLATION	endothelial	-0.009898	0.8615	0.9016
GSE57338	HF	REACTOME_TRNA_AMINOACYLATION	hepatocyte	-0.0325	0.5667	0.6624
GSE57338	HF	REACTOME_TRNA_AMINOACYLATION	cardiomyocyte	0.04009	0.4797	0.5757
GSE57338	HF	...	immune_pan	0.06571	0.2464	0.3214
GSE57338	HF	...	myeloid	0.07354	0.1944	0.2707
GSE57338	HF	...	fibroblast	-0.154	0.00635	0.01299
GSE57338	HF	...	endothelial	0.0408	0.472	0.574
GSE57338	HF	...	hepatocyte	0.08521	0.1325	0.2021
GSE57338	HF	...	cardiomyocyte	0.08734	0.1231	0.191
GSE57338	HF	REACTOME_RRNA_PROCESSING	immune_pan	0.2109	0.0001705	0.000465
GSE57338	HF	REACTOME_RRNA_PROCESSING	myeloid	0.1992	0.0003929	0.00104

S27 Table

Translation score gene accounting. For every cohort: the number of Tier A sets measured, the number of genes available for each module-derived score, and the fraction of each module present, replacing v1's single "227 genes" figure, which did not hold across cohorts.

Complete table: Table_S27_TATS_gene_accounting.csv (8 rows x 15 columns).

cohort	disease	tech	n.samples	n.tierA.sets	n.tierA expected	genes.HF module	genes.HF present	frac.HF present	genes.HCC module	genes.HCC present	frac.HCC present	HF.in.tierA	HCC.in.tierA	shared.HF.HCC
GSE57338	HF	array	313	15	15	153	153	1	146	146	1	73	26	5
GSE141198	HCC	RNA-seq	148	15	15	153	153	1	146	146	1	73	26	5
TCGA_LIHC	HCC	RNA-seq	421	15	15	153	152	0.993	146	142	0.973	73	25	5
GSE116250	HF	RNA-seq	64	15	15	153	145	0.948	146	131	0.897	72	22	5
GSE14520_GPL3921	HCC	array	445	14	15	153	103	0.673	146	78	0.534	62	13	2
GSE14520_GPL571	HCC	array	41	15	15	153	97	0.634	146	91	0.623	53	18	3
GSE76427	HCC	array	167	14	15	153	135	0.882	146	116	0.795	71	20	5
GSE141910	HF	RNA-seq	366	15	15	153	152	0.993	146	143	0.979	72	25	5

S28 Table (page 1 of 2)

Translation score concordance. Pairwise correlations among the canonical translation score, the HF module eigengene score, and the HCC module eigengene score, in each cohort, showing whether the choice of module determines the result.

cohort	score.a	score.b	r
GSE57338	TCS	ME_HF_weighted	0.6358
GSE57338	TCS	ME_HCC_weighted	0.7157
GSE57338	ME_HF_weighted	ME_HCC_weighted	0.3084
GSE141198	TCS	ME_HF_weighted	0.4629
GSE141198	TCS	ME_HCC_weighted	0.549
GSE141198	ME_HF_weighted	ME_HCC_weighted	0.7729
TCGA_LIHC	TCS	ME_HF_weighted	0.6019
TCGA_LIHC	TCS	ME_HCC_weighted	0.4723
TCGA_LIHC	ME_HF_weighted	ME_HCC_weighted	0.9115
GSE116250	TCS	ME_HF_weighted	0.2654
GSE116250	TCS	ME_HCC_weighted	0.4233
GSE116250	ME_HF_weighted	ME_HCC_weighted	0.8728
GSE14520_GPL3921	TCS	ME_HF_weighted	0.7852
GSE14520_GPL3921	TCS	ME_HCC_weighted	0.7032
GSE14520_GPL3921	ME_HF_weighted	ME_HCC_weighted	0.7274
GSE14520_GPL571	TCS	ME_HF_weighted	0.9041
GSE14520_GPL571	TCS	ME_HCC_weighted	0.8523
GSE14520_GPL571	ME_HF_weighted	ME_HCC_weighted	0.8041
GSE76427	TCS	ME_HF_weighted	0.8034
GSE76427	TCS	ME_HCC_weighted	0.6864
GSE76427	ME_HF_weighted	ME_HCC_weighted	0.6881
GSE141910	TCS	ME_HF_weighted	0.874

S28 Table (page 2 of 2)

Translation score concordance. Pairwise correlations among the canonical translation score, the HF module eigengene score, and the HCC module eigengene score, in each cohort, showing whether the choice of module determines the result.

Complete table: Table_S28_TATS_score_concordance.csv (24 rows x 4 columns).

cohort	score.a	score.b	r
GSE141910	TCS	ME_HCC_weighted	0.9171
GSE141910	ME_HF_weighted	ME_HCC_weighted	0.9506

S29 Table

Translation score effect sizes. Disease-versus-control effect size, P-value, and BH FDR for each score in each cohort.

Complete table: Table_S29_TATS_effect_sizes.csv (21 rows x 10 columns).

cohort	disease	score	n.disease	n.control	mean.disease	mean.control	hedges.g	p	p.fdr
GSE57338	HF	TCS	177	136	-0.1628	0.2119	-0.5692	1.99e-06	2.984e-06
GSE57338	HF	ME_HF_weighted	177	136	-0.1482	0.1929	-0.5526	3.905e-06	3.905e-06
GSE57338	HF	ME_HCC_weighted	177	136	-0.1088	0.1416	-0.6611	1.684e-08	5.053e-08
TCGA_LIHC	HCC	TCS	371	50	0.08138	-0.6039	0.8716	7.331e-15	7.331e-15
TCGA_LIHC	HCC	ME_HF_weighted	371	50	0.1042	-0.7732	1.356	1.003e-16	1.504e-16
TCGA_LIHC	HCC	ME_HCC_weighted	371	50	0.1102	-0.8174	1.433	2.134e-17	6.401e-17
GSE116250	HF	TCS	50	14	-0.023	0.08214	-0.1605	0.6378	0.9707
GSE116250	HF	ME_HF_weighted	50	14	0.02091	-0.07468	0.1541	0.7059	0.9707
GSE116250	HF	ME_HCC_weighted	50	14	-0.001622	0.005792	-0.01573	0.9707	0.9707
GSE14520_GPL3921	HCC	TCS	225	220	0.5148	-0.5265	2.043	8.106e-69	1.216e-68
GSE14520_GPL3921	HCC	ME_HF_weighted	225	220	0.3837	-0.3924	1.604	4.988e-50	4.988e-50
GSE14520_GPL3921	HCC	ME_HCC_weighted	225	220	0.3996	-0.4087	2.044	3.73e-71	1.119e-70
GSE14520_GPL571	HCC	TCS	22	19	0.3556	-0.4118	1.101	0.0008482	0.0008482
GSE14520_GPL571	HCC	ME_HF_weighted	22	19	0.3193	-0.3697	1.363	7.423e-05	0.0001113
GSE14520_GPL571	HCC	ME_HCC_weighted	22	19	0.3032	-0.3511	1.521	1.324e-05	3.972e-05
GSE76427	HCC	TCS	115	52	0.3121	-0.6902	1.512	1.888e-13	1.888e-13
GSE76427	HCC	ME_HF_weighted	115	52	0.1922	-0.425	1.765	1.209e-20	1.814e-20
GSE76427	HCC	ME_HCC_weighted	115	52	0.1628	-0.36	1.886	1.82e-23	5.459e-23
GSE141910	HF	TCS	200	166	0.04731	-0.057	0.1278	0.2178	0.6535
GSE141910	HF	ME_HF_weighted	200	166	0.02843	-0.03426	0.07868	0.4529	0.6793
GSE141910	HF	ME_HCC_weighted	200	166	-7.632e-05	9.196e-05	-0.000208	0.9984	0.9984

S30 Table (columns 1 of 2)

Transcription factor association. For every candidate transcription factor, in every cohort and for every score: the unadjusted Spearman correlation, the partial correlation adjusting for composition, both P-values and BH FDRs, the covariates used, whether the direction survived adjustment, and a flag marking factors that are themselves members of translation gene sets. The 582 rows are 14 factors x 3 scores (TCS and both module eigengene projections) across the arms available in each cohort — 312 rows on the all-samples arm and 270 on the disease-samples-only arm, the latter absent for GSE141198, which has no control arm. Because the table is too large to print in full, the printed supplement shows a bounded preview; the machine-readable CSV carries all 582.

Printed from Table_S30_TF_translation_association.csv (582 rows x 15 columns): 22 of 582 rows. Rows cited in the manuscript are printed first. The CSV is the complete and authoritative version.

					sets						
TCGA_LIHC	HCC	all	TCS	ATF4	False	421	0.2852	2.534e-09	0.2366	9.152e-07	...
TCGA_LIHC	HCC	all	TCS	MYC	False	421	0.1826	0.0001654	0.2327	1.385e-06	...
TCGA_LIHC	HCC	all	TCS	EIF4EBP1	True	421	0.2203	5.043e-06	0.1629	0.0007934	...
TCGA_LIHC	HCC	all	TCS	EIF2AK3	False	421	-0.3961	2.887e-17	-0.4518	1.458e-22	...
TCGA_LIHC	HCC	all	TCS	ERN1	False	421	-0.395	3.589e-17	-0.4153	5.562e-19	...
TCGA_LIHC	HCC	all	TCS	DDIT3	False	421	0.252	1.612e-07	0.1935	6.461e-05	...
TCGA_LIHC	HCC	all	TCS	MTOR	False	421	-0.3112	6.607e-11	-0.3879	1.437e-16	...
TCGA_LIHC	HCC	disease_only	TCS	ATF4	False	371	0.2469	1.484e-06	0.2457	1.666e-06	...
TCGA_LIHC	HCC	disease_only	TCS	MYC	False	371	0.2633	2.671e-07	0.2725	9.751e-08	...
TCGA_LIHC	HCC	disease_only	TCS	EIF4EBP1	True	371	0.1775	0.0005913	0.1752	0.0007023	...
TCGA_LIHC	HCC	disease_only	TCS	EIF2AK3	False	371	-0.4422	3.384e-19	-0.4619	5.307e-21	...
TCGA_LIHC	HCC	disease_only	TCS	ERN1	False	371	-0.42	2.745e-17	-0.4287	5.103e-18	...
TCGA_LIHC	HCC	disease_only	TCS	DDIT3	False	371	0.1692	0.001069	0.166	0.001331	...
TCGA_LIHC	HCC	disease_only	TCS	MTOR	False	371	-0.39	6.357e-15	-0.4093	2.03e-16	...
GSE57338	HF	all	TCS	ATF4	False	313	0.2842	3.148e-07	0.3065	3.136e-08	...
GSE57338	HF	all	TCS	MYC	False	313	0.1836	0.001105	0.2673	1.606e-06	...
GSE57338	HF	all	TCS	XBP1	False	313	0.3127	1.579e-08	0.3283	2.662e-09	...
GSE57338	HF	all	TCS	ATF6	False	313	0.1873	0.0008708	0.1918	0.0006455	...
GSE57338	HF	all	TCS	EIF4EBP1	True	313	0.3549	1.007e-10	0.3806	3.173e-12	...
GSE57338	HF	all	TCS	EIF2AK3	False	313	0.196	0.0004862	0.2081	0.0002092	...
GSE57338	HF	all	TCS	ERN1	False	313	-0.0123	0.8284	0.03411	0.5477	...
GSE57338	HF	all	TCS	DDIT3	False	313	0.2763	6.857e-07	0.2708	1.151e-06	...

S30 Table (columns 2 of 2)

Transcription factor association. For every candidate transcription factor, in every cohort and for every score: the unadjusted Spearman correlation, the partial correlation adjusting for composition, both P-values and BH FDRs, the covariates used, whether the direction survived adjustment, and a flag marking factors that are themselves members of translation gene sets. The 582 rows are 14 factors x 3 scores (TCS and both module eigengene projections) across the arms available in each cohort — 312 rows on the all-samples arm and 270 on the disease-samples-only arm, the latter absent for GSE141198, which has no control arm. Because the table is too large to print in full, the printed supplement shows a bounded preview; the machine-readable CSV carries all 582.

Printed from Table_S30_TF_translation_association.csv (582 rows x 15 columns): 22 of 582 rows. Rows cited in the manuscript are printed first. The CSV is the complete and authoritative version.

	unadjusted	adjusted	kept
TCGA_LIHC	8.87e-09	1.602e-06	True
TCGA_LIHC	0.0002573	2.154e-06	True
TCGA_LIHC	8.826e-06	0.0009256	True
TCGA_LIHC	2.512e-16	2.041e-21	True
TCGA_LIHC	2.512e-16	3.894e-18	True
TCGA_LIHC	4.512e-07	8.223e-05	True
TCGA_LIHC	3.083e-10	6.704e-16	True
TCGA_LIHC	2.967e-06	2.591e-06	True
TCGA_LIHC	9.348e-07	1.706e-07	True
TCGA_LIHC	0.0007526	0.0008938	True
TCGA_LIHC	4.738e-18	7.43e-20	True
TCGA_LIHC	1.922e-16	3.572e-17	True
TCGA_LIHC	0.001247	0.001434	True
TCGA_LIHC	2.966e-14	9.473e-16	True
GSE57338	1.01e-06	1.019e-07	True
GSE57338	0.001437	2.609e-06	True
GSE57338	6.842e-08	1.153e-08	True
GSE57338	0.001258	0.0008392	True
GSE57338	1.31e-09	2.062e-11	True
GSE57338	0.0007901	0.0003022	True
GSE57338	0.8284	0.5477	False
GSE57338	1.486e-06	2.138e-06	True

S31 Table

Gene significance threshold comparison. Hub gene counts under the v1 threshold (|GS| > 0.20), a stricter |GS| > 0.50, and the FDR < 0.05 criterion used in v2, in both networks. The three criteria are nested on these data: every gene passing |GS| > 0.50 also passes FDR < 0.05. |GS| > 0.50 is the strictest of the three and FDR < 0.05 the most permissive, so FDR < 0.05 was adopted for its calibration to sample size rather than because it is the most conservative.

Complete table: Table_S31_gene_significance_threshold.csv (2 rows x 9 columns).

cohort	module	n.module genes	n.present	n.absGS.gt 020	n.absGS.gt 050	n.absGS.gt 050.and.FDR05	n.FDR05.alone	median.abs.GS
GSE57338	purple	153	153	76	9	9	107	0.1969
TCGA_LIHC	magenta	146	142	119	2	2	133	0.3115

S32 Table

Module-internal robustness. Mean intra-modular topological overlap for HF purple and HCC magenta against both the uniform (v1-style) and connectivity-matched nulls, with Z, empirical P, and the observed-to-null mean ratio — the interpretable effect size, since Z is not comparable between nulls. Also reports the module and universe connectivity distributions that motivate the matched null, and jackknife stability as both the CV of the mean intra-modular connectivity and the median per-gene CV.

Complete table: Table_S32_module_internal_robustness.csv (2 rows x 23 columns).

disease	module	n.genes	power	n.samples	k.universe mean	k.module.mean	obs.mean intra.TOM	unmatched null.mean	unmatched null.sd	obs.over unmatched null	Z.unmatched	P.unmatched	matched.null mean	matched.null sd	obs.over matched.null	Z.matched	P.matched	n.perm	n.bins	jackknife.CV of.mean.kIM pct	jackknife median pergene.CV pct	jackknife.n degenerate
HF	purple	153	14	177	15.2	6.6	0.007868	0.002745	0.000372	2.87	13.77	0.0001	0.0007144	1.591e-05	11.01	449.7	0.0001	10000	20	2.66	53.28	0
HCC	magenta	146	14	148	74.1	21.1	0.01798	0.01834	0.002483	0.98	-0.14	0.544	0.003839	0.000161	4.68	87.83	0.0001	10000	20	2.48	49.68	0

S33 Table

Cross-disease module coherence. Mean pairwise correlation of each designated module's genes measured in the other disease's expression matrix, against a uniform null, for both modules.

Complete table: Table_S33_cross_disease_module_coherence.csv (2 rows x 11 columns).

module disease	module	tested.in	n.genes module	n.genes present	obs.mean pairwise.r	null.mean.r	null.sd.r	Z	p	n.perm
HF	purple	HCC	153	153	0.3256	0.02396	0.01357	22.22	0.000999	1000
HCC	magenta	HF	146	146	0.14	0.0355	0.007714	13.54	0.000999	1000

S34 Table (columns 1 of 2)

GSE89377 stage-spectrum effect sizes. Hedges g with bootstrap confidence intervals for every scored set at every stage of the hepatocarcinogenesis series (normal, chronic hepatitis low and high grade, cirrhosis, dysplastic nodule low and high grade, early HCC, HCC) against the normal arm, with stage index and group sizes.

Printed from Table_S34_GSE89377_stage_spectrum_effect_sizes.csv (280 rows x 16 columns): 22 of 280 rows. The CSV is the complete and authoritative version.

cohort	set	tier	stage	stage.index	n.stage	n.control	mean.stage	mean.control	hedges.g	ci.lo	ci.hi
GSE89377	KEGG_RIBOSOME	A	CH_low	2	8	13	0.7163	0.7098	0.6077	-0.08557	1.916
GSE89377	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	CH_low	2	8	13	0.06561	0.05359	0.3383	-0.5753	1.033
GSE89377	REACTOME_TRANSLATION	A	CH_low	2	8	13	0.5199	0.5105	0.7938	-0.004214	1.721
GSE89377	...	A	CH_low	2	8	13	0.6606	0.6541	0.6228	-0.06794	1.948
GSE89377	...	A	CH_low	2	8	13	0.7053	0.6994	0.5482	-0.1416	2.034
GSE89377	REACTOME_TRNA_AMINOACYLATION	A	CH_low	2	8	13	0.2042	0.1922	0.5245	-0.3154	1.288
GSE89377	...	A	CH_low	2	8	13	0.4885	0.4743	0.8748	0.1561	2.049
GSE89377	REACTOME_RRNA_PROCESSING	A	CH_low	2	8	13	0.4501	0.4465	0.263	-0.5074	1.314
GSE89377	...	A	CH_low	2	8	13	0.6718	0.6646	0.7225	-0.009904	1.879
GSE89377	...	A	CH_low	2	8	13	0.6202	0.6143	0.5784	-0.1496	1.849
GSE89377	...	A	CH_low	2	8	13	0.5656	0.5604	0.4891	-0.2639	1.826
GSE89377	REACTOME_MITOCHONDRIAL_TRANSLATION	A	CH_low	2	8	13	0.3839	0.3679	0.5962	-0.1771	1.313
GSE89377	...	A	CH_low	2	8	13	0.3891	0.3729	0.5872	-0.2008	1.308
GSE89377	...	A	CH_low	2	8	13	0.1115	0.1006	0.3482	-0.6116	1.033
GSE89377	...	A	CH_low	2	8	13	0.07167	0.05214	0.4909	-0.1715	1.164
GSE89377	REACTOME_TRNA_PROCESSING	B_extra	CH_low	2	8	13	0.03944	0.02821	0.8166	0.03496	2.489
GSE89377	...	B_extra	CH_low	2	8	13	0.07017	0.06093	0.6498	-0.08597	1.742
GSE89377	...	B_extra	CH_low	2	8	13	-0.01782	-0.02846	0.5359	-0.2434	1.771
GSE89377	...	B_extra	CH_low	2	8	13	0.3194	0.3212	-0.07267	-0.8216	0.8662
GSE89377	...	B_extra	CH_low	2	8	13	0.1534	0.1304	0.8239	0.06907	1.785
GSE89377	...	C_v1	CH_low	2	8	13	-0.01304	0.004659	-0.5278	-1.368	0.2648
GSE89377	HALLMARK_MYC_TARGETS_V1	C_v1	CH_low	2	8	13	0.442	0.4352	0.6546	-0.09191	1.762

S34 Table (columns 2 of 2)

GSE89377 stage-spectrum effect sizes. Hedges g with bootstrap confidence intervals for every scored set at every stage of the hepatocarcinogenesis series (normal, chronic hepatitis low and high grade, cirrhosis, dysplastic nodule low and high grade, early HCC, HCC) against the normal arm, with stage index and group sizes.

Printed from Table_S34_GSE89377_stage_spectrum_effect_sizes.csv (280 rows x 16 columns): 22 of 280 rows. The CSV is the complete and authoritative version.

cohort	p.ttest	ci.excludes.0	meets.min effect	p.fdr
GSE89377	0.1136	False	True	0.2587
GSE89377	0.3935	False	True	0.4497
GSE89377	0.08365	False	True	0.2587
GSE89377	0.1032	False	True	0.2587
GSE89377	0.1511	False	True	0.2587
GSE89377	0.2025	False	True	0.2822
GSE89377	0.04588	True	True	0.2587
GSE89377	0.5138	False	True	0.5709
GSE89377	0.07258	False	True	0.2587
GSE89377	0.1539	False	True	0.2587
GSE89377	0.2402	False	True	0.3099
GSE89377	0.1588	False	True	0.2587
GSE89377	0.1617	False	True	0.2587
GSE89377	0.3794	False	True	0.4497
GSE89377	0.194	False	True	0.2822
GSE89377	0.06312	True	True	0.2587
GSE89377	0.1297	False	True	0.2587
GSE89377	0.208	False	True	0.2822
GSE89377	0.8581	False	False	0.861
GSE89377	0.05394	True	True	0.2587
GSE89377	0.2117	False	True	0.2822
GSE89377	0.129	False	True	0.2587

S35 Table (page 1 of 2)

GSE89377 cirrhosis versus HCC specificity. Per-set effect sizes for the cirrhosis and HCC arms of GSE89377, the direction-agreement flag, and the cirrhosis-to-HCC magnitude ratio, for all scored sets with the tier assignment. The Tier A summary gives 12 of 15 concordant sets (rho = +0.668), with the three discordant sets being the aminoacyl-tRNA and mitochondrial ones.

set	tier	CH.low	CH.high	Cirrhosis	DN.low	DN.high	eHCC	HCC	cirrhosis same direction.as HCC	cirrhosis magnitude ratio
KEGG_RIBOSOME	A	0.6077	1.401	2.026	1.069	0.1532	0.1955	0.473	True	4.28
KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	0.3383	0.1583	-0.7455	0.2835	1.056	0.5937	0.2163	False	3.45
REACTOME_TRANSLATION	A	0.7938	1.41	1.048	1.271	1.245	1.047	0.6593	True	1.59
...	A	0.6228	1.546	1.87	1.166	0.387	0.3176	0.5826	True	3.21
...	A	0.5482	1.329	1.984	1.067	0.2267	0.3449	0.6257	True	3.17
REACTOME_TRNA_AMINOACYLATION	A	0.5245	0.4906	-0.6031	0.4622	1.316	0.645	0.4745	False	1.27
...	A	0.8748	1.599	0.9596	0.5227	1.457	0.4667	0.63	True	1.52
REACTOME_RRNA_PROCESSING	A	0.263	1.285	0.4994	0.1009	0.2918	-0.05144	0.4584	True	1.09
...	A	0.7225	1.57	2.135	1.323	0.457	0.6338	0.8066	True	2.65
...	A	0.5784	1.487	1.759	1.106	0.7364	1.025	0.9308	True	1.89
...	A	0.4891	1.352	1.587	0.9361	0.5535	0.821	0.8586	True	1.85
REACTOME_MITOCHONDRIAL_TRANSLATION	A	0.5962	0.8414	0.333	0.9075	1.064	0.6626	0.2156	True	1.54
...	A	0.5872	0.8517	0.4116	0.9415	1.067	0.7103	0.2528	True	1.63
...	A	0.3482	0.09756	-0.8501	0.3185	0.9819	0.5673	0.3475	False	2.45
...	A	0.4909	0.5692	-0.1247	0.4445	1.334	0.4034	-0.1284	True	0.97
REACTOME_TRNA_PROCESSING	B_extra	0.8166	0.7528	-0.6661	-0.04026	1.42	0.9236	0.9264	False	0.72
...	B_extra	0.6498	0.4546	-0.2671	-0.5052	1.016	0.7574	1.07	False	0.25
...	B_extra	0.5359	0.7989	-0.7863	0.1382	0.9917	0.8682	0.6945	False	1.13
...	B_extra	-0.07267	0.7461	-0.7096	-0.3476	-0.06897	-0.3248	0.2054	False	3.45
...	B_extra	0.8239	0.9141	-0.8385	0.8644	1.114	0.3061	-0.08342	True	10.05
...	C_v1	-0.5278	0.4688	0.4579	0.8314	0.7157	0.607	1.259	True	0.36
HALLMARK_MYC_TARGETS_V1	C_v1	0.6546	1.262	0.8349	0.2166	0.7321	0.6522	1.086	True	0.77

S35 Table (page 2 of 2)

GSE89377 cirrhosis versus HCC specificity. Per-set effect sizes for the cirrhosis and HCC arms of GSE89377, the direction-agreement flag, and the cirrhosis-to-HCC magnitude ratio, for all scored sets with the tier assignment. The Tier A summary gives 12 of 15 concordant sets (rho = +0.668), with the three discordant sets being the aminoacyl-tRNA and mitochondrial ones.

Complete table: Table_S35_GSE89377_HCC_vs_cirrhosis_specificity.csv (40 rows x 11 columns).

set	tier	CH.low	CH.high	Cirrhosis	DN.low	DN.high	eHCC	HCC	cirrhosis same direction.as HCC	cirrhosis magnitude ratio
...	C_v1	0.7333	0.8672	0.7617	0.7991	1.313	1.778	0.5512	True	1.38
...	C_v1	0.4735	0.9398	0.7481	1.377	1.996	2.038	0.8045	True	0.93
...	C_v1	1.075	1.25	2.447	1.442	1.281	2.402	1.512	True	1.62
HALLMARK_MYC_TARGETS_V2	C_v1	-0.1527	0.3537	-0.9576	-0.5578	0.07352	-0.3329	0.0639	False	14.99
...	C_v1	0.6815	0.5576	1.361	0.2749	0.1921	0.2325	0.8582	True	1.59
...	C_v1	0.5305	0.7123	0.7774	1.614	1.129	1.564	0.8789	True	0.88
...	C_v1	-0.7148	-0.2335	-1.175	0.1381	0.7672	0.583	0.1248	False	9.42
...	C_v1	0.6993	0.9831	1.773	1.046	0.634	1.017	0.7561	True	2.34
HALLMARK_MTORC1_SIGNALING	C_v1	0.1061	0.7573	0.2183	0.3418	0.1831	0.4828	0.1424	True	1.53
...	C_v1	-0.7476	-0.1563	0.5314	-0.1604	-0.5909	-0.8128	-0.3968	False	1.34
...	C_v1	-0.4784	-1.339	-1.694	-0.7458	-0.5584	-0.9661	-0.5042	True	3.36
...	C_v1	0.6033	-0.5445	0.4805	0.7025	-0.2016	0.1729	0.1385	True	3.47
...	C_v1	-0.06783	-0.06755	0.7144	-0.4492	-0.5026	-0.4036	-0.0294	False	24.3
...	C_v1	-1.193	-1.534	-1.328	-0.4481	-0.2499	0.14	0.5933	False	2.24
...	C_v1	-0.6065	-0.7608	-1.276	-0.2282	0.7444	0.5299	0.005175	False	246.5
...	C_v1	-0.8277	-1.623	-1.765	-0.4436	-0.2199	-0.213	0.2767	False	6.38
...	C_v1	-0.7832	-1.526	-1.743	-0.3871	0.0005128	-0.05953	0.446	False	3.91
...	C_v1	-0.7832	-1.526	-1.743	-0.3871	0.0005128	-0.05953	0.446	False	3.91

S36 Table

Within-disease connectivity-free module cohesion. Mean pairwise Pearson correlation among each designated module's genes measured in that module's own expression matrix, against a uniform random gene-set null, with Z and empirical P. Reported alongside the module-to-universe connectivity ratio, which is the confound this measure is free of. This is the within-disease counterpart to S33, and it is the quantity that shows the connectivity-matched null to be the correct one: under the connectivity-normalised TOM null the HCC module scores Z = -0.14 (P = 0.54), whereas this unnormalised measure of the same property scores Z = 26.00.

Complete table: Table_S36_within_disease_cohesion.csv (2 rows x 14 columns).

disease	module	n.genes	n.samples	n.genes universe	obs.mean pairwise.r	null.mean.r	null.sd.r	Z	p	n.perm	k.module.mean	k.universe mean	k.module.over universe
HF	purple	153	177	10213	0.3403	0.03552	0.007442	40.96	0.000999	1000	6.6	15.2	0.434
HCC	magenta	146	148	10213	0.3916	0.02393	0.01414	26	0.000999	1000	21.1	74.1	0.285

S37 Table (page 1 of 3)

Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	set	tier	n	g.unadjusted	g.adjusted	attenuation	survives
GSE116250	...	A	64	-1.046	-1.033	0.01252	True
GSE116250	REACTOME_MITOCHONDRIAL_TRANSLATION	A	64	-0.9057	-0.8895	0.0178	True
GSE116250	...	A	64	-0.8454	-0.8515	-0.007179	True
GSE116250	...	A	64	-0.7373	-0.714	0.03163	True
GSE116250	REACTOME_TRANSLATION	A	64	-0.4471	-0.435	0.02705	False
GSE116250	...	A	64	0.4269	0.448	-0.04932	False
GSE116250	...	A	64	0.4125	0.4064	0.01484	False
GSE116250	...	A	64	0.3671	0.3535	0.03703	False
GSE116250	...	A	64	0.3085	0.3024	0.01956	False
GSE116250	KEGG_RIBOSOME	A	64	0.2264	0.2188	0.03327	False
GSE116250	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	64	0.2204	0.2538	-0.1517	False
GSE116250	...	A	64	0.2126	0.2093	0.01526	False
GSE116250	REACTOME_TRNA_AMINOACYLATION	A	64	0.1057	0.1367	-0.2929	False
GSE116250	...	A	64	-0.04968	-0.06322	-0.2727	False
GSE116250	REACTOME_RRNA_PROCESSING	A	64	-0.02782	-0.02653	0.04659	False
GSE141910	...	A	366	0.5115	0.5163	-0.009307	True
GSE141910	...	A	366	-0.4446	-0.4473	-0.006014	True
GSE141910	...	A	366	0.3338	0.3185	0.04587	True
GSE141910	KEGG_RIBOSOME	A	366	0.3292	0.3136	0.04746	True
GSE141910	...	A	366	0.2276	0.2176	0.04379	True
GSE141910	...	A	366	-0.2201	-0.2072	0.05859	False
GSE141910	...	A	366	0.2038	0.1929	0.05385	False

S37 Table (page 2 of 3)

Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

cohort	set	tier	n	g.unadjusted	g.adjusted	attenuation	survives
GSE141910	...	A	366	0.1844	0.1789	0.02941	False
GSE141910	...	A	366	0.1155	0.1434	-0.2409	False
GSE141910	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	366	0.1064	0.1039	0.02357	False
GSE141910	...	A	366	0.08926	0.1115	-0.2489	False
GSE141910	REACTOME_TRANSLATION	A	366	0.06832	0.0853	-0.2485	False
GSE141910	REACTOME_MITOCHONDRIAL_TRANSLATION	A	366	0.06534	0.09329	-0.4278	False
GSE141910	REACTOME_TRNA_AMINOACYLATION	A	366	0.02325	0.02391	-0.02828	False
GSE141910	REACTOME_RRNA_PROCESSING	A	366	-0.01464	-0.01653	-0.1286	False
GSE57338	...	A	313	-0.9199	-0.8337	0.09374	True
GSE57338	REACTOME_RRNA_PROCESSING	A	313	-0.8114	-0.6811	0.1605	True
GSE57338	...	A	313	-0.6511	-0.4672	0.2824	True
GSE57338	...	A	313	-0.6071	-0.4302	0.2914	True
GSE57338	...	A	313	-0.5993	-0.4254	0.2902	True
GSE57338	...	A	313	0.5582	0.4634	0.1698	True
GSE57338	...	A	313	-0.5521	-0.4165	0.2456	True
GSE57338	REACTOME_TRANSLATION	A	313	-0.539	-0.5068	0.05964	True
GSE57338	...	A	313	-0.4757	-0.2831	0.4048	True
GSE57338	KEGG_RIBOSOME	A	313	-0.4473	-0.2565	0.4264	True
GSE57338	REACTOME_TRNA_AMINOACYLATION	A	313	-0.3621	-0.3498	0.0338	True
GSE57338	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	313	-0.2085	-0.2122	-0.01773	False
GSE57338	...	A	313	-0.1005	-0.2159	-1.148	False
GSE57338	REACTOME_MITOCHONDRIAL_TRANSLATION	A	313	-0.08824	-0.1916	-1.171	False

S37 Table (page 3 of 3)

Age and sex adjustment. For every scored set in the four cohorts whose public annotation carries both age and sex (GSE57338, GSE116250, GSE141910, TCGA-LIHC): the disease-versus-control effect standardised as β/σ with and without the two covariates in the model, both bootstrap confidence intervals, both P-values and BH FDRs, the standardised age and sex coefficients, the maximum VIF of the fit, and whether the set keeps its sign and its gate status. Adjusted and unadjusted fits are computed on the same complete-case sample set.

Printed from Table_S37_clinical_covariate_adjustment.csv (163 rows x 25 columns): 8 of 25 columns and 60 of 163 rows. Rows cited in the manuscript are printed first. The CSV is the complete and authoritative version.

cohort	set	tier	n	g.unadjusted	g.adjusted	attenuation	survives
GSE57338	...	A	313	-0.08121	-0.1826	-1.249	False
TCGA_LIHC	...	A	420	1.173	1.16	0.01137	True
TCGA_LIHC	REACTOME_TRNA_AMINOACYLATION	A	420	1.119	1.116	0.002185	True
TCGA_LIHC	...	A	420	1.101	1.069	0.02893	True
TCGA_LIHC	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS	A	420	1.024	1.023	0.0006651	True
TCGA_LIHC	...	A	420	0.9665	0.9262	0.04177	True
TCGA_LIHC	...	A	420	0.8786	0.8433	0.04013	True
TCGA_LIHC	...	A	420	0.8552	0.8176	0.04396	True
TCGA_LIHC	REACTOME_RRNA_PROCESSING	A	420	0.8332	0.7992	0.04081	True
TCGA_LIHC	...	A	420	0.7321	0.695	0.05069	True
TCGA_LIHC	REACTOME_TRANSLATION	A	420	0.6849	0.6595	0.03714	True
TCGA_LIHC	...	A	420	0.6592	0.661	-0.002746	True
TCGA_LIHC	KEGG_RIBOSOME	A	420	0.6375	0.5961	0.06502	True
TCGA_LIHC	...	A	420	-0.2566	-0.2902	-0.131	False
TCGA_LIHC	...	A	420	0.1763	0.1604	0.09003	False
TCGA_LIHC	REACTOME_MITOCHONDRIAL_TRANSLATION	A	420	0.1235	0.107	0.1336	False

S38 Table

Mirror recheck after age and sex adjustment. Cross-disease effect-size correlation and mirror count between TCGA-LIHC and each heart-failure cohort, computed on unadjusted and on age- and sex-adjusted effects under an identical mirror gate, for Tier A sets and for all scored sets. The Tier A mirror count is unchanged in every cohort; across all scored sets it is unchanged everywhere except GSE141910, where it falls from 15 to 12. The discovery cohort's correlation moves from rho = -0.550 to rho = -0.461.

Complete table: Table_S38_age_sex_mirror_recheck.csv (6 rows x 8 columns).

hf.cohort	tier	n.sets	rho unadjusted	mirror unadjusted	rho.adjusted	mirror adjusted	gate
GSE116250	A	15	0.2107	1	0.275	1	opposite signs, |g| >= 0.2, CI...
GSE116250	ALL	40	0.2859	7	0.2938	7	opposite signs, |g| >= 0.2, CI...
GSE141910	A	15	-0.1464	1	-0.1214	1	opposite signs, |g| >= 0.2, CI...
GSE141910	ALL	41	-0.08589	15	-0.07265	12	opposite signs, |g| >= 0.2, CI...
GSE57338	A	15	-0.55	10	-0.4607	10	opposite signs, |g| >= 0.2, CI...
GSE57338	ALL	41	-0.4563	26	-0.3929	26	opposite signs, |g| >= 0.2, CI...

S39 Table (panel a) (columns 1 of 2)

Translation score and overall survival. (a) For every HCC cohort with usable outcome annotation: the Cox hazard ratio per IQR of the score and as a median split, with 95% confidence interval, Cox and log-rank P-values, the proportional-hazards test, and BH FDR, before and after adjustment for tissue composition. Each row names the covariates used — hepatocyte (the tumour-purity proxy), proliferation, and immune content — and reports the covariate hazard ratios and a flag for whether the proportional-hazards assumption holds. Models are fitted only where at least ten events per covariate are available, and the sample count and event count are given for every row, so a cohort absent from a given covariate set can be identified as underpowered rather than omitted. (b) Survival annotation coverage: for every cohort in the pipeline, whether a survival field is deposited, how many samples join it, how many are dropped for a missing or non-positive time, and how many events remain, with the source of each annotation. The three heart-failure cohorts deposit no survival field and are therefore not analysed, and GSE14520_GPL571 carries annotation whose sample identifiers do not intersect that platform's cohort, so it is reported as not analysed rather than as covered.

Printed from Table_S39_translation_score_survival.csv (49 rows x 19 columns): 22 of 49 rows. Rows cited in the manuscript are printed first. The CSV is the complete and authoritative version.

GSE141198	NBNC	TCS	continuous_per_IQR	none	0	19	10	1.303	0.6893	2.464	0.4151
GSE141198		TCS	continuous_per_IQR	none	0	148	94	1.038	0.7774	1.385	0.8014
GSE141198		TCS	continuous_per_IQR	proliferation	1	148	94	1.011	0.7551	1.353	0.943
GSE141198		TCS	continuous_per_IQR	purity	1	148	94	1.026	0.7655	1.376	0.8616
GSE141198		TCS	continuous_per_IQR	purity+proliferation+immune	3	148	94	1.016	0.7531	1.372	0.9155
GSE14520_GPL3921		TCS	continuous_per_IQR	none	0	221	85	1.548	1.174	2.041	0.001971
GSE14520_GPL3921		TCS	continuous_per_IQR	proliferation	1	221	85	1.465	1.094	1.963	0.01038
GSE14520_GPL3921		TCS	continuous_per_IQR	purity	1	221	85	1.519	1.145	2.017	0.003809
GSE14520_GPL3921		TCS	continuous_per_IQR	purity+proliferation+immune	3	221	85	1.431	1.062	1.928	0.01846
GSE76427		TCS	continuous_per_IQR	none	0	115	23	1.353	0.761	2.405	0.3032
GSE76427		TCS	continuous_per_IQR	proliferation	1	115	23	1.366	0.7673	2.432	0.2891
GSE76427		TCS	continuous_per_IQR	purity	1	115	23	1.35	0.7619	2.391	0.304
TCGA_LIHC		TCS	continuous_per_IQR	none	0	366	130	1.13	0.9011	1.417	0.2897
TCGA_LIHC		TCS	continuous_per_IQR	proliferation	1	366	130	1.127	0.8988	1.413	0.3007
TCGA_LIHC		TCS	continuous_per_IQR	purity	1	366	130	1.146	0.9153	1.434	0.2351
TCGA_LIHC		TCS	continuous_per_IQR	purity+proliferation+immune	3	366	130	1.131	0.9027	1.418	0.2841
GSE141198		ME_HCC_weighted	continuous_per_IQR	none	0	148	94	0.8801	0.6557	1.181	0.3951
GSE141198		ME_HCC_weighted	median_split	none	0	148	94	0.9453	0.6301	1.418	0.786
GSE141198		ME_HF_weighted	continuous_per_IQR	none	0	148	94	0.8675	0.6406	1.175	0.3583
GSE141198		ME_HF_weighted	median_split	none	0	148	94	0.7681	0.5114	1.153	0.2034

S39 Table (panel a) (columns 2 of 2)

Translation score and overall survival. (a) For every HCC cohort with usable outcome annotation: the Cox hazard ratio per IQR of the score and as a median split, with 95% confidence interval, Cox and log-rank P-values, the proportional-hazards test, and BH FDR, before and after adjustment for tissue composition. Each row names the covariates used — hepatocyte (the tumour-purity proxy), proliferation, and immune content — and reports the covariate hazard ratios and a flag for whether the proportional-hazards assumption holds. Models are fitted only where at least ten events per covariate are available, and the sample count and event count are given for every row, so a cohort absent from a given covariate set can be identified as underpowered rather than omitted. (b) Survival annotation coverage: for every cohort in the pipeline, whether a survival field is deposited, how many samples join it, how many are dropped for a missing or non-positive time, and how many events remain, with the source of each annotation. The three heart-failure cohorts deposit no survival field and are therefore not analysed, and GSE14520_GPL571 carries annotation whose sample identifiers do not intersect that platform's cohort, so it is reported as not analysed rather than as covered.

Printed from Table_S39_translation_score_survival.csv (49 rows x 19 columns): 22 of 49 rows. Rows cited in the manuscript are printed first. The CSV is the complete and authoritative version.

GSE141198		0.2061	True	1		days	0.8746
GSE141198		0.1317	True	1		days	0.943
GSE141198		0.3329	True	1.012	proliferation 1.09	days	0.943
GSE141198		0.3133	True	1.008	hepatocyte 0.97	days	0.943
GSE141198		0.6645	True	1.15	hepatocyte 1.00; proliferation...	days	0.943
GSE14520_GPL3921		0.4857	True	1		months	0.02285
GSE14520_GPL3921		0.7262	True	1.13	proliferation 1.14	months	0.02628
GSE14520_GPL3921		0.7029	True	1.03	hepatocyte 0.95	months	0.02285
GSE14520_GPL3921		0.1974	True	1.151	hepatocyte 0.96; proliferation...	months	0.03165
GSE76427		0.7987	True	1		years	0.5067
GSE76427		0.9041	True	1.024	proliferation 0.93	years	0.5067
GSE76427		0.2229	True	1.002	hepatocyte 1.11	years	0.5067
TCGA_LIHC		0.433	True	1		days	0.3007
TCGA_LIHC		0.000949	False	1.004	proliferation 1.47	days	0.3007
TCGA_LIHC		0.771	True	1	hepatocyte 0.86	days	0.3007
TCGA_LIHC		0.01151	False	1.082	hepatocyte 0.89; proliferation...	days	0.3007
GSE141198		0.3492	True	1		days	0.8746
GSE141198	0.7875	0.2122	True	1		days	0.943
GSE141198		0.3955	True	1		days	0.8746
GSE141198	0.2015	0.9521	True	1		days	0.8746

S39 Table (panel b)

Translation score and overall survival. (a) For every HCC cohort with usable outcome annotation: the Cox hazard ratio per IQR of the score and as a median split, with 95% confidence interval, Cox and log-rank P-values, the proportional-hazards test, and BH FDR, before and after adjustment for tissue composition. Each row names the covariates used — hepatocyte (the tumour-purity proxy), proliferation, and immune content — and reports the covariate hazard ratios and a flag for whether the proportional-hazards assumption holds. Models are fitted only where at least ten events per covariate are available, and the sample count and event count are given for every row, so a cohort absent from a given covariate set can be identified as underpowered rather than omitted. (b) Survival annotation coverage: for every cohort in the pipeline, whether a survival field is deposited, how many samples join it, how many are dropped for a missing or non-positive time, and how many events remain, with the source of each annotation. The three heart-failure cohorts deposit no survival field and are therefore not analysed, and GSE14520_GPL571 carries annotation whose sample identifiers do not intersect that platform's cohort, so it is reported as not analysed rather than as covered.

Complete table: Table_S39b_survival_annotation_coverage.csv (8 rows x 11 columns).

GSE141198	HCC	True	148	148	0	148	94	days		GSE141198_clinical.rds:...
TCGA_LIHC	HCC	True	421	371	5	366	130	days		TCGA_LIHC_se.rds colData:...
GSE116250	HF	False	64	0	0	0	0		no survival field deposited in...	none deposited
GSE14520_GPL3921	HCC	True	445	225	4	221	85	months		...
GSE14520_GPL571	HCC	False	41	0	0	0	0	months	annotation present but no...	...
GSE76427	HCC	True	167	167	52	115	23	years		GSE76427_series_matrix.txt.gz:...
GSE141910	HF	False	366	0	0	0	0		no survival field deposited in...	none deposited

S40 Table (panel a) (page 1 of 2)

Paired-sample sensitivity analyses. (a) Sample-level pairing manifest for all 1,965 analysed samples across the eight cohorts, with patient keys, pair completeness, inclusion flags, and exclusion reasons; unmatched samples are marked and excluded from paired statistics. (b) Pairing summary per cohort, including complete pairs and unmatched disease/control counts. (c) Per-set paired effect sizes for the Tier A sets available in each of the four paired cohorts: number of pairs analysed, within-pair mean difference, standard deviation, dz, Jp, gpaired, bootstrap 95% confidence interval, two-sided within-pair sign-flip permutation P, within-cohort Benjamini–Hochberg FDR, the corresponding full-cohort Hedges g, the complete-pair-subset unpaired g, and direction agreement. (d) Cohort-level summary of pair counts, Tier A sets tested, direction agreement, confidence-interval support, nominal and FDR counts. (e) Statistical definitions and resampling parameters. The complete CSVs are supplied as machine-readable files; the PDF prints bounded previews of panels a and c, and the CSVs are the authoritative full text.

cohort	group	patient.key	paired.design	included.in paired	pair.id	exclusion reason	sample.id
TCGA_LIHC	disease	TCGA-EP-A3RK	True	True	TCGA_LIHC:TCGA-EP-A3RK		TCGA-EP-A3RK-01A-11R-A22L-07
TCGA_LIHC	control	TCGA-EP-A3RK	True	True	TCGA_LIHC:TCGA-EP-A3RK		TCGA-EP-A3RK-11A-11R-A22L-07
TCGA_LIHC	disease	TCGA-BC-A10Y	True	True	TCGA_LIHC:TCGA-BC-A10Y		TCGA-BC-A10Y-01A-11R-A131-07
TCGA_LIHC	disease	TCGA-BC-A10Z	True	True	TCGA_LIHC:TCGA-BC-A10Z		TCGA-BC-A10Z-01A-11R-A131-07
TCGA_LIHC	control	TCGA-DD-A3A1	True	True	TCGA_LIHC:TCGA-DD-A3A1		TCGA-DD-A3A1-11A-11R-A213-07
TCGA_LIHC	disease	TCGA-DD-A1EH	True	True	TCGA_LIHC:TCGA-DD-A1EH		TCGA-DD-A1EH-01A-11R-A131-07
TCGA_LIHC	control	TCGA-FV-A2QR	True	True	TCGA_LIHC:TCGA-FV-A2QR		TCGA-FV-A2QR-11A-11R-A213-07
TCGA_LIHC	control	TCGA-DD-A39V	True	True	TCGA_LIHC:TCGA-DD-A39V		TCGA-DD-A39V-11A-11R-A213-07
TCGA_LIHC	control	TCGA-DD-A11C	True	True	TCGA_LIHC:TCGA-DD-A11C		TCGA-DD-A11C-11A-11R-A131-07
TCGA_LIHC	disease	TCGA-FV-A2QR	True	True	TCGA_LIHC:TCGA-FV-A2QR		TCGA-FV-A2QR-01A-11R-A213-07
TCGA_LIHC	control	TCGA-DD-A3A5	True	True	TCGA_LIHC:TCGA-DD-A3A5		TCGA-DD-A3A5-11A-11R-A22L-07
TCGA_LIHC	disease	TCGA-DD-A3A5	True	True	TCGA_LIHC:TCGA-DD-A3A5		TCGA-DD-A3A5-01A-11R-A22L-07
TCGA_LIHC	control	TCGA-FV-A23B	True	True	TCGA_LIHC:TCGA-FV-A23B		TCGA-FV-A23B-11A-11R-A16W-07
TCGA_LIHC	control	TCGA-BC-A10U	True	True	TCGA_LIHC:TCGA-BC-A10U		TCGA-BC-A10U-11A-11R-A131-07
TCGA_LIHC	control	TCGA-DD-A3A6	True	True	TCGA_LIHC:TCGA-DD-A3A6		TCGA-DD-A3A6-11A-11R-A22L-07
TCGA_LIHC	disease	TCGA-FV-A3I1	True	True	TCGA_LIHC:TCGA-FV-A3I1		TCGA-FV-A3I1-01A-11R-A22L-07
TCGA_LIHC	disease	TCGA-DD-A1EB	True	True	TCGA_LIHC:TCGA-DD-A1EB		TCGA-DD-A1EB-01A-11R-A131-07
TCGA_LIHC	control	TCGA-DD-A1EB	True	True	TCGA_LIHC:TCGA-DD-A1EB		TCGA-DD-A1EB-11A-11R-A131-07
TCGA_LIHC	disease	TCGA-DD-A3A6	True	True	TCGA_LIHC:TCGA-DD-A3A6		TCGA-DD-A3A6-01A-11R-A22L-07
TCGA_LIHC	control	TCGA-FV-A3I1	True	True	TCGA_LIHC:TCGA-FV-A3I1		TCGA-FV-A3I1-11A-11R-A22L-07
TCGA_LIHC	control	TCGA-DD-A39X	True	True	TCGA_LIHC:TCGA-DD-A39X		TCGA-DD-A39X-11A-11R-A213-07
TCGA_LIHC	disease	TCGA-FV-A23B	True	True	TCGA_LIHC:TCGA-FV-A23B		TCGA-FV-A23B-01A-11R-A16W-07

S40 Table (panel a) (page 2 of 2)

Paired-sample sensitivity analyses. (a) Sample-level pairing manifest for all 1,965 analysed samples across the eight cohorts, with patient keys, pair completeness, inclusion flags, and exclusion reasons; unmatched samples are marked and excluded from paired statistics. (b) Pairing summary per cohort, including complete pairs and unmatched disease/control counts. (c) Per-set paired effect sizes for the Tier A sets available in each of the four paired cohorts: number of pairs analysed, within-pair mean difference, standard deviation, dz, Jp, gpaired, bootstrap 95% confidence interval, two-sided within-pair sign-flip permutation P, within-cohort Benjamini–Hochberg FDR, the corresponding full-cohort Hedges g, the complete-pair-subset unpaired g, and direction agreement. (d) Cohort-level summary of pair counts, Tier A sets tested, direction agreement, confidence-interval support, nominal and FDR counts. (e) Statistical definitions and resampling parameters. The complete CSVs are supplied as machine-readable files; the PDF prints bounded previews of panels a and c, and the CSVs are the authoritative full text.

Printed from Table_S40a_paired_sample_manifest.csv (1965 rows x 10 columns): 8 of 10 columns and 44 of 1965 rows. The CSV is the complete and authoritative version.

TCGA_LIHC	disease	TCGA-BC-A10U	True	True	TCGA_LIHC:TCGA-BC-A10U		TCGA-BC-A10U-01A-11R-A131-07
TCGA_LIHC	control	TCGA-DD-A114	True	True	TCGA_LIHC:TCGA-DD-A114		TCGA-DD-A114-11A-12R-A131-07
TCGA_LIHC	control	TCGA-DD-A1EC	True	True	TCGA_LIHC:TCGA-DD-A1EC		TCGA-DD-A1EC-11A-11R-A131-07
TCGA_LIHC	control	TCGA-BC-A110	True	True	TCGA_LIHC:TCGA-BC-A110		TCGA-BC-A110-11A-11R-A131-07
TCGA_LIHC	disease	TCGA-BC-A110	True	True	TCGA_LIHC:TCGA-BC-A110		TCGA-BC-A110-01A-11R-A131-07
TCGA_LIHC	disease	TCGA-DD-A39X	True	True	TCGA_LIHC:TCGA-DD-A39X		TCGA-DD-A39X-01A-11R-A213-07
TCGA_LIHC	disease	TCGA-DD-A1EC	True	True	TCGA_LIHC:TCGA-DD-A1EC		TCGA-DD-A1EC-01A-21R-A131-07
TCGA_LIHC	disease	TCGA-DD-A11C	True	True	TCGA_LIHC:TCGA-DD-A11C		TCGA-DD-A11C-01A-11R-A131-07
TCGA_LIHC	control	TCGA-BD-A2L6	True	True	TCGA_LIHC:TCGA-BD-A2L6		TCGA-BD-A2L6-11A-21R-A213-07
TCGA_LIHC	control	TCGA-DD-A11B	True	True	TCGA_LIHC:TCGA-DD-A11B		TCGA-DD-A11B-11A-11R-A131-07
TCGA_LIHC	disease	TCGA-G3-A3CH	True	True	TCGA_LIHC:TCGA-G3-A3CH		TCGA-G3-A3CH-01A-11R-A22L-07
TCGA_LIHC	control	TCGA-DD-A1EI	True	True	TCGA_LIHC:TCGA-DD-A1EI		TCGA-DD-A1EI-11A-11R-A131-07
TCGA_LIHC	disease	TCGA-DD-A39Z	True	True	TCGA_LIHC:TCGA-DD-A39Z		TCGA-DD-A39Z-01A-11R-A213-07
TCGA_LIHC	control	TCGA-DD-A39W	True	True	TCGA_LIHC:TCGA-DD-A39W		TCGA-DD-A39W-11A-11R-A213-07
TCGA_LIHC	disease	TCGA-DD-A39W	True	True	TCGA_LIHC:TCGA-DD-A39W		TCGA-DD-A39W-01A-11R-A213-07
TCGA_LIHC	control	TCGA-BC-A216	True	True	TCGA_LIHC:TCGA-BC-A216		TCGA-BC-A216-11A-11R-A155-07
TCGA_LIHC	disease	TCGA-DD-A3A3	True	True	TCGA_LIHC:TCGA-DD-A3A3		TCGA-DD-A3A3-01A-11R-A22L-07
TCGA_LIHC	control	TCGA-DD-A3A3	True	True	TCGA_LIHC:TCGA-DD-A3A3		TCGA-DD-A3A3-11A-11R-A22L-07
TCGA_LIHC	control	TCGA-DD-A119	True	True	TCGA_LIHC:TCGA-DD-A119		TCGA-DD-A119-11A-11R-A131-07
TCGA_LIHC	control	TCGA-DD-A1EG	True	True	TCGA_LIHC:TCGA-DD-A1EG		TCGA-DD-A1EG-11A-11R-A213-07
TCGA_LIHC	disease	TCGA-DD-A118	True	True	TCGA_LIHC:TCGA-DD-A118		TCGA-DD-A118-01A-11R-A131-07

S40 Table (panel b)

Paired-sample sensitivity analyses. (a) Sample-level pairing manifest for all 1,965 analysed samples across the eight cohorts, with patient keys, pair completeness, inclusion flags, and exclusion reasons; unmatched samples are marked and excluded from paired statistics. (b) Pairing summary per cohort, including complete pairs and unmatched disease/control counts. (c) Per-set paired effect sizes for the Tier A sets available in each of the four paired cohorts: number of pairs analysed, within-pair mean difference, standard deviation, dz, Jp, gpaired, bootstrap 95% confidence interval, two-sided within-pair sign-flip permutation P, within-cohort Benjamini–Hochberg FDR, the corresponding full-cohort Hedges g, the complete-pair-subset unpaired g, and direction agreement. (d) Cohort-level summary of pair counts, Tier A sets tested, direction agreement, confidence-interval support, nominal and FDR counts. (e) Statistical definitions and resampling parameters. The complete CSVs are supplied as machine-readable files; the PDF prints bounded previews of panels a and c, and the CSVs are the authoritative full text.

Complete table: Table_S40b_pairing_summary.csv (8 rows x 13 columns).

GSE37338	HF	False	313	177	136		0			0	313	none (unpaired case-control)
GSE141198	HCC	False	148	148	0		0			0	148	none (tumour-only, no control arm)
TCGA_LIHC	HCC	True	421	371	50	371	50	321	0	100	321	barcode chars 1-12; type chars...
GSE116250	HF	False	64	50	14		0			0	64	none (unpaired case-control)
GSE14520_GPL3921	HCC	True	445	225	220	232	213	12	7	426	19	Individual stripped of trailing...
GSE14520_GPL571	HCC	True	41	22	19	22	19	3	0	38	3	Individual stripped of trailing...
GSE76427	HCC	True	167	115	52	115	52	63	0	104	63	patient.id
GSE141910	HF	False	366	200	166		0			0	366	none (unpaired case-control)

S40 Table (panel c) (page 1 of 2)

Paired-sample sensitivity analyses. (a) Sample-level pairing manifest for all 1,965 analysed samples across the eight cohorts, with patient keys, pair completeness, inclusion flags, and exclusion reasons; unmatched samples are marked and excluded from paired statistics. (b) Pairing summary per cohort, including complete pairs and unmatched disease/control counts. (c) Per-set paired effect sizes for the Tier A sets available in each of the four paired cohorts: number of pairs analysed, within-pair mean difference, standard deviation, dz, Jp, gpaired, bootstrap 95% confidence interval, two-sided within-pair sign-flip permutation P, within-cohort Benjamini–Hochberg FDR, the corresponding full-cohort Hedges g, the complete-pair-subset unpaired g, and direction agreement. (d) Cohort-level summary of pair counts, Tier A sets tested, direction agreement, confidence-interval support, nominal and FDR counts. (e) Statistical definitions and resampling parameters. The complete CSVs are supplied as machine-readable files; the PDF prints bounded previews of panels a and c, and the CSVs are the authoritative full text.

cohort	n.pairs analysed	g.paired.z	paired.ci.lo	paired.ci.hi	p.perm.paired	p.fdr.within cohort	hedges.g.full cohort	direction matches.full cohort	set
GSE14520_GPL3921	213	1.02	0.8702	1.194	9.999e-05	9.999e-05	1.348	True	KEGG_RIBOSOME
GSE14520_GPL3921	213	0.4503	0.3168	0.5955	9.999e-05	9.999e-05	0.5985	True	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS
GSE14520_GPL3921	213	1.776	1.543	2.084	9.999e-05	9.999e-05	2.34	True	REACTOME_TRANSLATION
GSE14520_GPL3921	213	1.182	1.039	1.354	9.999e-05	9.999e-05	1.599	True	...
GSE14520_GPL3921	213	0.7962	0.662	0.9571	9.999e-05	9.999e-05	1.07	True	...
GSE14520_GPL3921	213	1.019	0.8693	1.19	9.999e-05	9.999e-05	1.392	True	REACTOME_TRNA_AMINOACYLATION
GSE14520_GPL3921	213	1.284	1.134	1.464	9.999e-05	9.999e-05	1.71	True	REACTOME_RRNA_PROCESSING
GSE14520_GPL3921	213	1.424	1.188	1.768	9.999e-05	9.999e-05	1.885	True	...
GSE14520_GPL3921	213	1.845	1.583	2.165	9.999e-05	9.999e-05	2.355	True	...
GSE14520_GPL3921	213	1.944	1.677	2.277	9.999e-05	9.999e-05	2.499	True	...
GSE14520_GPL3921	213	1.064	0.881	1.274	9.999e-05	9.999e-05	1.371	True	REACTOME_MITOCHONDRIAL_TRANSLATION
GSE14520_GPL3921	213	1.051	0.8712	1.261	9.999e-05	9.999e-05	1.355	True	...
GSE14520_GPL3921	213	0.4503	0.3168	0.5955	9.999e-05	9.999e-05	0.5985	True	...
GSE14520_GPL3921	213	-0.4737	-0.6222	-0.3331	9.999e-05	9.999e-05	-0.5951	True	...
GSE14520_GPL571	19	1.124	0.7794	1.67	9.999e-05	0.0015	1.244	True	REACTOME_RRNA_PROCESSING
GSE14520_GPL571	19	0.8454	0.3682	1.743	0.0015	0.01025	1.29	True	...
GSE14520_GPL571	19	0.8358	0.3163	1.917	0.0021	0.01025	1.247	True	...
GSE14520_GPL571	19	0.7713	0.2352	1.902	0.0037	0.01025	0.9839	True	...
GSE14520_GPL571	19	0.7787	0.2597	1.837	0.0039	0.01025	1.105	True	...
GSE14520_GPL571	19	0.7775	0.2647	1.79	0.0041	0.01025	1.203	True	REACTOME_TRANSLATION
GSE14520_GPL571	19	0.7368	0.1986	1.96	0.0049	0.0105	0.9826	True	...
GSE14520_GPL571	19	0.5192	0.05258	1.443	0.0295	0.0469	0.6263	True	KEGG_RIBOSOME

S40 Table (panel c) (page 2 of 2)

Paired-sample sensitivity analyses. (a) Sample-level pairing manifest for all 1,965 analysed samples across the eight cohorts, with patient keys, pair completeness, inclusion flags, and exclusion reasons; unmatched samples are marked and excluded from paired statistics. (b) Pairing summary per cohort, including complete pairs and unmatched disease/control counts. (c) Per-set paired effect sizes for the Tier A sets available in each of the four paired cohorts: number of pairs analysed, within-pair mean difference, standard deviation, dz, Jp, gpaired, bootstrap 95% confidence interval, two-sided within-pair sign-flip permutation P, within-cohort Benjamini–Hochberg FDR, the corresponding full-cohort Hedges g, the complete-pair-subset unpaired g, and direction agreement. (d) Cohort-level summary of pair counts, Tier A sets tested, direction agreement, confidence-interval support, nominal and FDR counts. (e) Statistical definitions and resampling parameters. The complete CSVs are supplied as machine-readable files; the PDF prints bounded previews of panels a and c, and the CSVs are the authoritative full text.

Printed from Table_S40c_paired_effect_sizes.csv (58 rows x 26 columns): 10 of 26 columns and 44 of 58 rows. The CSV is the complete and authoritative version.

GSE14520_GPL571	19	0.5136	0.07444	1.088	0.0304	0.0469	0.8371	True	REACTOME_MITOCHONDRIAL_TRANSLATION
GSE14520_GPL571	19	0.5047	0.07351	1.039	0.032	0.0469	0.8308	True	...
GSE14520_GPL571	19	0.4793	0.07613	0.9655	0.0344	0.0469	0.6662	True	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS
GSE14520_GPL571	19	0.4327	-0.01274	1.083	0.05869	0.06992	0.6957	True	REACTOME_TRNA_AMINOACYLATION
GSE14520_GPL571	19	0.4419	-0.01115	1.331	0.06059	0.06992	0.444	True	...
GSE14520_GPL571	19	0.2493	-0.1859	0.6849	0.2784	0.2983	0.3713	True	...
GSE14520_GPL571	19	0.0355	-0.4221	0.4946	0.8743	0.8743	0.05438	True	...
GSE76427	52	0.8774	0.5715	1.276	9.999e-05	0.0001077	1.013	True	KEGG_RIBOSOME
GSE76427	52	0.8528	0.5601	1.293	9.999e-05	0.0001077	1.042	True	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS
GSE76427	52	1.164	0.8195	1.651	9.999e-05	0.0001077	1.603	True	REACTOME_TRANSLATION
GSE76427	52	1.029	0.7188	1.427	9.999e-05	0.0001077	1.206	True	...
GSE76427	52	1.071	0.7407	1.516	9.999e-05	0.0001077	1.189	True	...
GSE76427	52	1.039	0.6954	1.576	9.999e-05	0.0001077	1.377	True	REACTOME_TRNA_AMINOACYLATION
GSE76427	52	1.179	0.8782	1.59	9.999e-05	0.0001077	1.265	True	REACTOME_RRNA_PROCESSING
GSE76427	52	1.09	0.7383	1.581	9.999e-05	0.0001077	1.341	True	...
GSE76427	52	1.145	0.771	1.684	9.999e-05	0.0001077	1.412	True	...
GSE76427	52	1.157	0.7723	1.754	9.999e-05	0.0001077	1.53	True	...
GSE76427	52	0.7697	0.5086	1.126	9.999e-05	0.0001077	1.081	True	REACTOME_MITOCHONDRIAL_TRANSLATION
GSE76427	52	0.7812	0.5187	1.131	9.999e-05	0.0001077	1.104	True	...
GSE76427	52	0.959	0.6447	1.42	9.999e-05	0.0001077	1.175	True	...
GSE76427	52	-0.1335	-0.4045	0.1215	0.3259	0.3259	-0.07637	True	...
TCGA_LIHC	50	1.088	0.7554	1.535	9.999e-05	0.00025	1.024	True	KEGG_AMINOACYL_TRNA_BIOSYNTHESIS

S40 Table (panel d)

Paired-sample sensitivity analyses. (a) Sample-level pairing manifest for all 1,965 analysed samples across the eight cohorts, with patient keys, pair completeness, inclusion flags, and exclusion reasons; unmatched samples are marked and excluded from paired statistics. (b) Pairing summary per cohort, including complete pairs and unmatched disease/control counts. (c) Per-set paired effect sizes for the Tier A sets available in each of the four paired cohorts: number of pairs analysed, within-pair mean difference, standard deviation, dz, Jp, gpaired, bootstrap 95% confidence interval, two-sided within-pair sign-flip permutation P, within-cohort Benjamini–Hochberg FDR, the corresponding full-cohort Hedges g, the complete-pair-subset unpaired g, and direction agreement. (d) Cohort-level summary of pair counts, Tier A sets tested, direction agreement, confidence-interval support, nominal and FDR counts. (e) Statistical definitions and resampling parameters. The complete CSVs are supplied as machine-readable files; the PDF prints bounded previews of panels a and c, and the CSVs are the authoritative full text.

Complete table: Table_S40d_paired_summary.csv (4 rows x 13 columns).

GSE14520_GPL3921	HCC	213	213	14	14	0	14	14	14	14	1.058	1.382
GSE14520_GPL571	HCC	19	19	15	15	0	15	11	11	11	0.5192	0.8371
GSE76427	HCC	52	52	14	14	0	14	13	13	13	1.034	1.198
TCGA_LIHC	HCC	50	50	15	15	0	15	11	11	11	0.5878	0.8342

S40 Table (panel e)

Paired-sample sensitivity analyses. (a) Sample-level pairing manifest for all 1,965 analysed samples across the eight cohorts, with patient keys, pair completeness, inclusion flags, and exclusion reasons; unmatched samples are marked and excluded from paired statistics. (b) Pairing summary per cohort, including complete pairs and unmatched disease/control counts. (c) Per-set paired effect sizes for the Tier A sets available in each of the four paired cohorts: number of pairs analysed, within-pair mean difference, standard deviation, dz, Jp, gpaired, bootstrap 95% confidence interval, two-sided within-pair sign-flip permutation P, within-cohort Benjamini–Hochberg FDR, the corresponding full-cohort Hedges g, the complete-pair-subset unpaired g, and direction agreement. (d) Cohort-level summary of pair counts, Tier A sets tested, direction agreement, confidence-interval support, nominal and FDR counts. (e) Statistical definitions and resampling parameters. The complete CSVs are supplied as machine-readable files; the PDF prints bounded previews of panels a and c, and the CSVs are the authoritative full text.

Complete table: Table_S40e_paired_methods.csv (16 rows x 2 columns).

paired_unit	complete patient pair (one...
effect_size	g_paired_z = J_p * mean(d_i) /...
standardisation_denominator	SD of within-pair differences...
hedges_correction	J_p = 1 - 3/(4*n_pairs - 5)
bootstrap_unit	patient pairs, with replacement
bootstrap_replicates	2000
bootstrap_CI	percentile 95%
permutation_scheme	within-pair disease/control...
shared_sign_matrix_across_sets	yes: one sign matrix per cohort...
permutation_replicates	10000
permutation_p	two-sided (1 + #{|stat_perm| >=...
multiplicity_family	Tier A sets within each paired...
fdr_method	BH
unmatched_samples	excluded from paired analysis;...
scope	complete-pair subset...
cross_disease_rho	not computed